

SGN – Assignment #1

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1 Periodic orbit

Exercise 1

Consider the 3D Earth–Moon Circular Restricted Three-Body Problem with $\mu = 0.012150$. Note that the CRTBP has an integral of motion, that is, the Jacobi constant

$$J(x, y, z, v_x, v_y, v_z) := 2\Omega(x, y, z) - v^2 = C$$

where $\Omega(x, y, z) = \frac{1}{2}(x^2 + y^2) + \frac{1-\mu}{r_1} + \frac{\mu}{r_2} + \frac{1}{2}\mu(1-\mu)$ and $v^2 = v_x^2 + v_y^2 + v_z^2$.

- 1) Find the coordinates of the five Lagrange points L_i in the rotating, adimensional reference frame with at least 10-digit accuracy and report their Jacobi constant C_i .

Solutions to the 3D CRTBP satisfy the symmetry

$$\mathcal{S} : (x, y, z, \dot{x}, \dot{y}, \dot{z}, t) \rightarrow (x, -y, z, -\dot{x}, \dot{y}, -\dot{z}, -t).$$

Thus, a trajectory that crosses perpendicularly the $y = 0$ plane twice is a periodic orbit.

- 2) Given the initial guess $\mathbf{x}_0 = (x_0, y_0, z_0, v_{x0}, v_{y0}, v_{z0})$, with

$$\begin{aligned} x_0 &= 1.068792441776 \\ y_0 &= 0 \\ z_0 &= 0.071093328515 \\ v_{x0} &= 0 \\ v_{y0} &= 0.319422926485 \\ v_{z0} &= 0 \end{aligned}$$

Find the periodic halo orbit having a Jacobi Constant $C = 3.09$; that is, develop the theoretical framework and implement a differential correction scheme that uses the STM, either approximated through finite differences **or** achieved by integrating the variational equation.

Hint: Consider working on $\varphi(\mathbf{x} + \Delta\mathbf{x}, t + \Delta t)$ and $J(\mathbf{x} + \Delta\mathbf{x})$ and then enforce perpendicular cross of $y = 0$ and Jacobi energy.

The periodic orbits in the CRTBP exist in families. These can be computed by ‘continuing’ the orbits along one coordinate or one parameter, e.g., the Jacobi energy C . The *numerical continuation* is an iterative process in which the desired variable is *gradually* varied, while the rest of the initial guess is taken from the solution of the previous iteration, thus aiding the convergence process.

- 3) By gradually decreasing C and using numerical continuation, compute the families of halo orbits until $C = 3.04$.

(8 points)

1.1 Lagrangian points

In the 3D Earth–Moon Circular Restricted Three-Body Problem (CRTBP), the Lagrangian points represent the equilibrium points of the system. Thus, they can be found as zeros of the gradient of the scalar potential $\Omega(x, y, z)$:

$$\nabla\Omega = \begin{bmatrix} \frac{\partial\Omega}{\partial x}(x, y, z) \\ \frac{\partial\Omega}{\partial y}(x, y, z) \\ \frac{\partial\Omega}{\partial z}(x, y, z) \end{bmatrix} = \begin{bmatrix} x - \frac{1-\mu}{r_1^3}(x+\mu) + \frac{\mu}{r_2^3}(1-\mu-x) \\ y - \frac{1-\mu}{r_1^3}y - \frac{\mu}{r_2^3}y \\ -\frac{1-\mu}{r_1^3}z - \frac{\mu}{r_2^3}z \end{bmatrix} \quad (1)$$

In particular, it can be shown that $\frac{\partial\Omega}{\partial z} = 0$ if and only if $z = 0$, reducing the problem to finding the stationary points of Ω in the x - y plane.

A zero-finding problem has been implemented using `fsolve` with a tolerance of 10^{-10} in order to obtain an accuracy solution of 10 digits. For the initial guesses, a contour plot of a scalar function representing the norm of the potential gradient has been used, as shown in Figure 1. As visible from the figure, the five zeros correspond to the five Lagrangian points: three of them are proved to be collinear (L_1 , L_2 and L_3) while the other two are called triangular points (L_4 and L_5).

Their coordinates with respect to Earth and Moon in an adimensional reference frame are presented in Table 1, along with their Jacobi constant C_i , computed setting to zero all velocity components since they are equilibrium points.

A second method to retrieve the Lagrangian points is described in Appendix A, exploiting mathematical derivations to check the proper use of `fsolve` and the results.

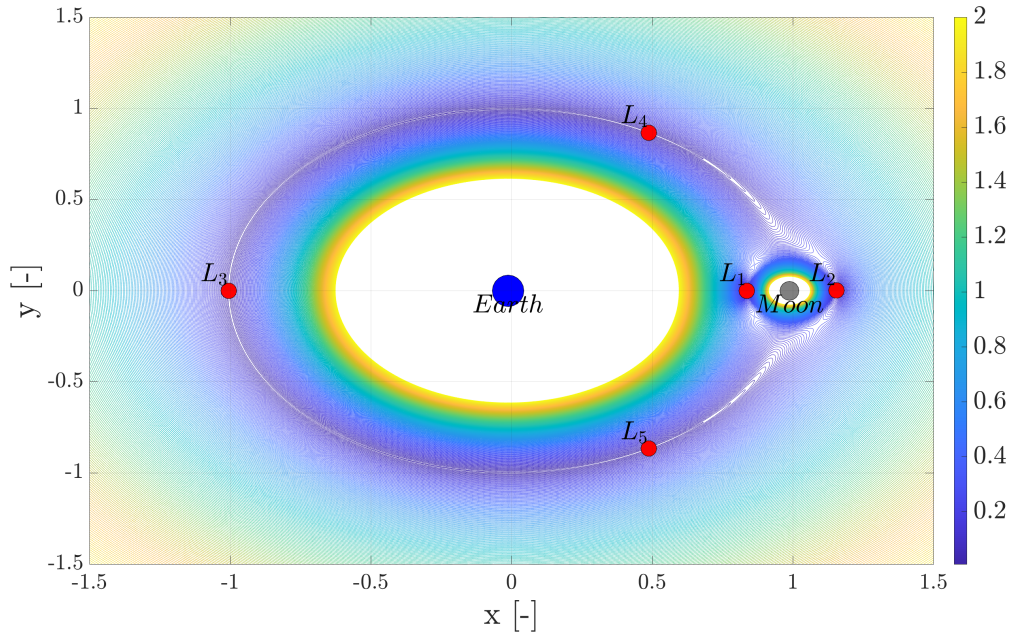


Figure 1: Contour plot and Lagrangian points in Earth–Moon adimensional rotating frame

	L_1	L_2	L_3	L_4	L_5
x	+0.8369180073	+1.1556799131	-1.0050624018	+0.4878500000	-0.4878500000
y	0.0000000000	0.0000000000	0.0000000000	+0.8660254038	-0.8660254038
C	+3.2003380950	+3.1841582164	+3.0241489429	+3.0000000000	+3.0000000000

Table 1: Lagrangian points coordinates and Jacobi constants

1.2 Halo orbit

Given the initial guess $\mathbf{x}_0$, a *while loop* is constructed in order to implement the differential correction needed to match the required Jacobi constant C . In each iteration of the loop, a propagation of the current state is performed up to an event time that corresponds to the crossing of the *xz-plane* (where $y = 0$), from which the deviation in the final state is computed. Now, to be able to retrieve the differential correction on current state and time, it is necessary to construct the augmented matrix derived from the first-order Taylor expansion of $\varphi(\mathbf{x} + \Delta\mathbf{x}, t + \Delta t)$ and $J(\mathbf{x} + \Delta\mathbf{x})$. In fact, the system to be solved has the following form:

$$\mathbf{A}_{(7 \times 7)}^* \Delta_{(7 \times 1)} = \mathbf{X}_{target(7 \times 1)} - \mathbf{X}_{real(7 \times 1)} = \mathbf{X}_{err(7 \times 1)} \quad (2)$$

where $\mathbf{A}^*$ is the augmented matrix, Δ is the differential correction that contains both state and time correction, and $\mathbf{X}_{err}$ is the error vector of state and Jacobi constant.

To develop this framework, it is necessary to linearize the ODE flow that governs the CRTBP dynamics (Appendix B) and the Jacobi constant expression with respect to the initial condition x_0 and the propagation time t_f :

$$\begin{cases} x(t) = \varphi(x_0 + \delta x_0, t_0, t_f + \delta t_f) \approx \varphi(x_0, t_0, t_f) + \frac{\partial \varphi(x_0, t_0, t_f)}{\partial x_0} \delta x_0 + \frac{\partial \varphi(x_0, t_0, t_f)}{\partial t_f} \delta t_f \\ C(x_0 + \delta x_0) \approx C(x_0) + \frac{\partial C(x_0)}{\partial x_0} \delta x_0 + \frac{\partial C(x_0)}{\partial t_f} \delta t_f \end{cases} \quad (3)$$

The matrix $\mathbf{A}^*$ results to be 7×7 and contains:

1. The entire State Transition Matrix $\Phi_{6 \times 6}(\mathbf{x}_0, t_0; t) = \frac{\partial \varphi}{\partial \mathbf{x}_0}(\mathbf{x}_0, t_0; t)$ (STM) computed integrating the variational approach (size 6×6).
2. The right-hand side (RHS) of the dynamics $\dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x})$ as the last column (6×1).
3. The Jacobian of J as the last row (1×6).
4. A zero in position 7-7 because the partial derivative of the Jacobi constant with respect to time is null.

Now, it is possible to simplify the system enforcing perpendicular cross of *xz-plane* ($y = 0$). In fact, it implies that no differential correction is needed for y , v_x and v_z , condition that translates into the elimination of columns 2, 4, 6 of $\mathbf{A}^*$. Furthermore, the target and real state for x , z and v_y are the same, a simplification that reduces the system down to 4 equations.

$$\begin{bmatrix} \Phi_{21} & \Phi_{23} & \Phi_{25} & v_y & \frac{\partial \Omega}{\partial x} \\ \Phi_{41} & \Phi_{43} & \Phi_{45} & 2v_y + \frac{\partial \Omega}{\partial x} & \frac{\partial \Omega}{\partial z} \\ \Phi_{61} & \Phi_{63} & \Phi_{65} & \frac{\partial \Omega}{\partial z} & 0 \\ 2\frac{\partial \Omega}{\partial x} & 2\frac{\partial \Omega}{\partial z} & -2v_y & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta z \\ \Delta v_y \\ \Delta t \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 3.09 \end{bmatrix} - \begin{bmatrix} y_f \\ v_{x,f} \\ v_{z,f} \\ C_f \end{bmatrix} \quad (4)$$

Finally, from Equation 4, it is possible to update the current state by adding the computed differential correction Δ . Algorithm 1 presents a summary of the iterative process just explained. This iterative process continues until convergence is achieved: once the value of the error vector reaches the selected tolerance, the loop ends and the correct initial state is retrieved in Table 2.

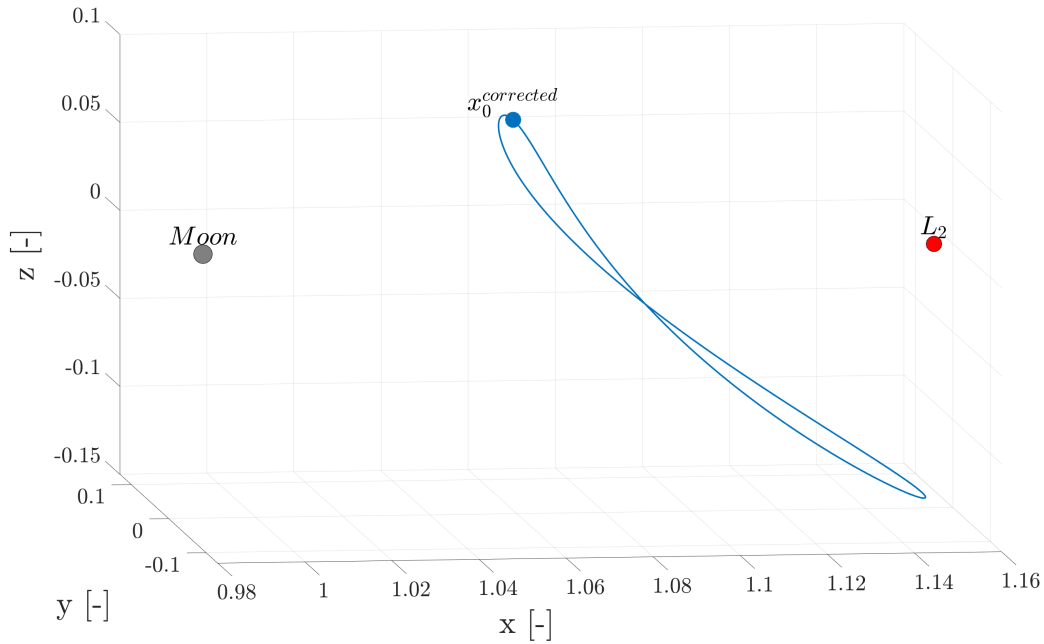
x	y	z
+1.0590402077	0.0000000000	+0.0739277378
v_x	v_y	v_z
0.0000000000	+0.3469245709	0.0000000000

Table 2: Corrected initial state of the halo orbit with $C = 3.09$

Algorithm 1 Find Halo Orbit Initial Conditions**Require:** Initial guess $\mathbf{x}_0^{(0)}$, $t_f^{(0)}$, C_{target} **Ensure:** Correct initial state $\mathbf{x}_0$ and final time t_f

- 1: Initialize $\text{iter} = 0$, errors $err_{y_f}^{(0)}$, $err_{v_{xf}}^{(0)}$, $err_{v_{zf}}^{(0)}$, $err_C^{(0)} = 1$, and tolerance $\text{tol} = 10^{-12}$
- 2: **while** $\max\left(\left|err_{y_f}^{(k)}\right|, \left|err_{v_{xf}}^{(k)}\right|, \left|err_{v_{zf}}^{(k)}\right|, \left|err_C^{(k)}\right|\right) > \text{tol}$ **and** $\text{iter} < N_{\text{max}}$ **do**
- 3: Propagate $\mathbf{x}_0^{(k)}$ up to $t_f^{(k)}$
- 4: Compute $\Phi(0, t_f^{(k)})$ and $\mathbf{J}'(\mathbf{x}_0^{(k)})$
- 5: Compute errors: $err_{y_f}^{(k)}$, $err_{v_{xf}}^{(k)}$, $err_{v_{zf}}^{(k)}$, $err_C^{(k)}$
- 6: Solve Equation 4 and find $\delta x_0^{(k)}$, $\delta z_0^{(k)}$, $\delta v_{y_0}^{(k)}$, $\delta t_f^{(k)}$
- 7: Update $\mathbf{x}_0^{(k+1)}$ and $t_f^{(k+1)}$ with corrections $\delta x_0^{(k)}$, $\delta z_0^{(k)}$, $\delta v_{y_0}^{(k)}$, $\delta t_f^{(k)}$
- 8: Update $k = k + 1$
- 9: **end while**

Figure 2 shows the final halo orbit derived from the propagation of the corrected initial state in the Earth–Moon adimensional rotating reference frame. It is evident that the orbit is periodic and around the Lagrangian point L_2 , as expected.

**Figure 2:** 3D halo orbit with $C = 3.09$ in the Earth–Moon adimensional rotating frame

1.3 Families of halo orbits

In order to obtain the correct initial state for the halo orbit of $C = 3.04$, the solution process described before is collected in a function (*find_Halo*), which is iterated in a *for loop*, as presented in Algorithm 2.

Algorithm 2 Numerical Continuation

Require: Initial guess $\mathbf{x}_0^{(0)}$ and C_{vect}

Ensure: Corrected initial state for each energy level C_{ref}

- 1: **for** each C_{ref} **do**
 - 2: Apply differential correction to compute $\mathbf{x}_0^k$ through function *find_Halo*
 - 3: Update initial state $\mathbf{x}_0^{(k+1)}$ for the next iteration with $\mathbf{x}_0^k$
 - 4: Propagate the trajectory using the corrected initial state $\mathbf{x}_0^k$ for the current step
 - 5: Plot the 3D orbit for the current step
 - 6: **end for**
-

Exploiting numerical continuation through a Jacobi constant step of $C = 0.01$, it is possible to picture all families of halo orbits (as shown in Figure 3), from which the correct initial state of the last orbit is recovered and reported in Table 3.

x	y	z
+1.0125655235	0.0000000000	+0.0672339583

v_x	v_y	v_z
0.0000000000	+0.5103251959	0.0000000000

Table 3: Corrected initial state of the halo orbit with $C = 3.04$

Figure 3 also shows how decreasing the Jacobi constant reference leads to larger periodic orbits which gradually drift away from L_2 .

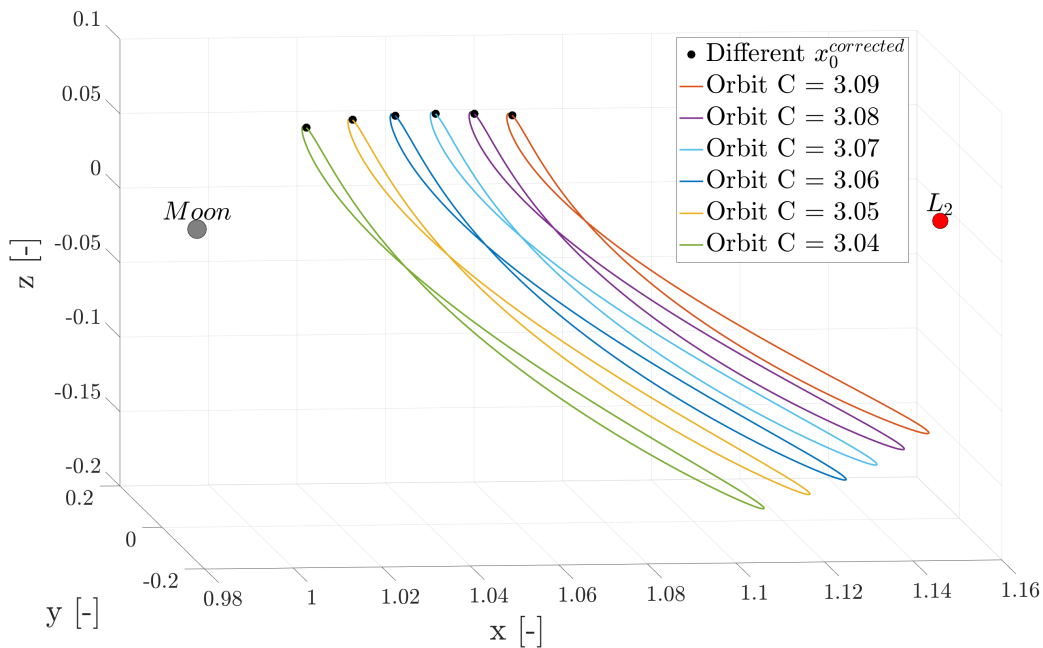


Figure 3: Families of halo orbits in the Earth–Moon adimensional rotating frame

2 Impulsive guidance

Exercise 2

Consider the two-impulse transfer problem stated in Section 3.1 (Topputo, 2013)*.

- 1) Using the procedure in Section 3.2, produce a first guess solution using $\alpha = 0.2\pi$, $\beta = 1.41$, $\delta = 4$, and $t_i = 2$. Plot the solution in both the rotating frame and Earth-centered inertial frame (see Appendix 1 in (Topputo, 2013)). Consider the parameters listed in Table 4 and extract the radius and gravitational parameters of the Earth and Moon from the provided kernels and use the latter to compute the parameter μ .

Symbol	Value	Units	Meaning
m_s	3.28900541×10^5	-	Scaled mass of the Sun
ρ	3.88811143×10^2	-	Scaled Sun-(Earth+Moon) distance
ω_s	$-9.25195985 \times 10^{-1}$	-	Scaled angular velocity of the Sun
ω_{em}	$2.66186135 \times 10^{-1}$	s^{-1}	Earth-Moon angular velocity
l_{em}	3.84405×10^8	m	Earth-Moon distance
h_i	167	km	Altitude of departure orbit
h_f	100	km	Altitude of arrival orbit
DU	3.84405000×10^5	km	Distance Unit
TU	4.34256461	days	Time Unit
VU	1.02454018	km/s	Velocity Unit

Table 4: Constants to be considered to solve the PBRFBP. The units of distance, time, and velocity are used to map scaled quantities into physical units.

- 2) Considering the first guess in 1) and using $\{\mathbf{x}_i, t_i, t_f\}$ as variables, solve the problem in Section 3.1 with simple shooting in the following cases
 - a) without providing any derivative to the solver, and
 - b) by providing the derivatives and by estimating the state transition matrix with variational equations.
- 3) Considering the first guess solution in 1) and the procedure in Section 3.3, solve the problem with multiple shooting taking $N = 4$ and using the variational equation to compute the Jacobian of the nonlinear equality constraints.
- 4) Perform an n-body propagation using the solution $\{\mathbf{x}_i, t_i, t_f\}$ obtained in point 2), transformed in Earth-centered inertial frame and into physical units. To move from 2-D to 3-D, assume that the position and velocity components in inertial frame are $r_z(t_i) = 0$ and $v_z(t_i) = 0$. To perform the propagation it is necessary to identify the epoch t_i . This can be done by mapping the relative position of the Earth, Moon and Sun in the PCRTBP to a similar condition in the real world:
 - a) Consider the definition of $\theta(t)$ provided in Section 2.2 to compute the angle $\theta_i = \theta(t_i)$. Note that this angle corresponds to the angle between the rotating frame x -axis, aligned to the position vector from the Earth-Moon System Barycenter (EMB) to the Moon, and the Sun direction.
 - b) The angle θ ranges between $[0, 2\pi]$ and it covers this domain in approximately the revolution period of the Moon around the Earth.

*F. Topputo, “On optimal two-impulse Earth–Moon transfers in a four-body model”, *Celestial Mechanics and Dynamical Astronomy*, Vol. 117, pp. 279–313, 2013, DOI: 10.1007/s10569-013-9513-8.

- c) Solve a zero-finding problem to determine the epoch at which the angle Moon-EMB-Sun is equal to θ_i , considering as starting epoch 2024 Sep 28 00:00:00.000 TDB.
Hints: Exploit the SPK kernels to define the orientation of the rotating frame axes in the inertial frame for an epoch t . Consider only the projection of the EMB-Sun position vector onto the so-defined x-y plane to compute the angle (planar motion).

Plot the propagated orbit and compare it to the previously found solutions.

(11 points)

2.1 First guess solution

In order to produce a first guess solution of a two-impulse Earth–Moon transfer, first it is necessary to extract the radius and gravitational parameters of the Earth and Moon from the kernels, through the Matlab tool `fcspace_bodvrdand`, and use the latter to compute the adimensional mass parameter $\mu = \frac{\mu_M}{\mu_E + \mu_M}$. Taking into account the oblateness of the Earth, the mean radius $r_E = \frac{r_{equatorial} + r_{polar}}{2}$ has been chosen for all the exercises.

Given the parameters in Table 4, it is possible to retrieve the initial guess state $\mathbf{x}_0$ in the Earth–Moon adimensional rotating frame as presented below and in Table 5.

$$\begin{cases} r_0 = \frac{r_E + h_i}{DU} \\ r_f = \frac{r_M + h_f}{DU} \end{cases} \implies \begin{cases} x_0 = r_0 \cos(\alpha) - \mu \\ y_0 = r_0 \sin(\alpha) \end{cases} \quad (5)$$

$$v_0 = \beta \sqrt{\frac{1 - \mu}{r_0}} \implies \begin{cases} v_{x_0} = -(v_0 - r_0) \sin(\alpha) \\ v_{y_0} = (v_0 - r_0) \cos(\alpha) \end{cases} \quad (6)$$

$$\mathbf{x}_0 = [x_0; y_0; v_{x_0}; v_{y_0}] \quad (7)$$

$r_{x,0}$ [DU]	$r_{y,0}$ [DU]	$v_{x,0}$ [VU]	$v_{y,0}$ [VU]
+0.001602	+0.009992	-6.307917	+8.682103

Table 5: Initial guess in Earth-Moon rotating frame

The computed $\mathbf{x}_0$ is then propagated from t_i to $t_f = t_i + \delta$ integrating the equation ruling the Planar Bicircular Restricted Four-Body Problem (PBRFBP) dynamics (Appendix C). The resulting orbit is exhibited in both the Earth–Moon rotating and Earth–centered inertial adimensional reference frames in Figure 4. Here and in general, a state $\mathbf{x}$ expressed in the Earth–Moon rotating frame can be converted to the Earth–centered inertial frame through the following transformation.

$$\begin{cases} x_{in} &= (x + \mu) \cos(t) - y \sin(t) \\ y_{in} &= (x + \mu) \sin(t) + y \cos(t) \\ v_{x_{in}} &= (v_x - y) \cos(t) - (v_y + x + \mu) \sin(t) \\ v_{y_{in}} &= (v_x - y) \sin(t) + (v_y + x + \mu) \cos(t) \end{cases} \quad (8)$$

$$\mathbf{x}_{in} = [x_{in}; y_{in}; v_{x_{in}}; v_{y_{in}}] \quad (9)$$

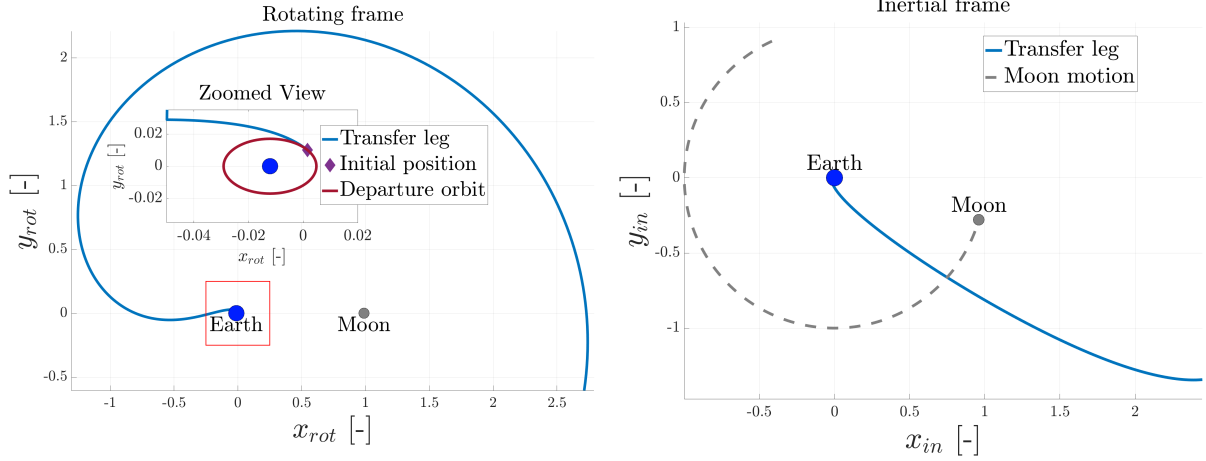


Figure 4: First guess solution in both adimensional frames

2.2 Simple shooting

Simple shooting is an optimization technique formulated as a Non-Linear Programming (NLP in Appendix D). Knowing that there are no linear equality constraints and no nonlinear inequality constraints, the problem is stated as:

$$\min_{\mathbf{y}} f_{obj}(\mathbf{y}) \quad \text{such that:} \quad \begin{cases} A_{ineq} \mathbf{y} \leq \mathbf{b}_{ineq} \\ \mathbf{c}_{eq}(\mathbf{y}) = 0 \\ \mathbf{y}^{LB} \leq \mathbf{y} \leq \mathbf{y}^{UB} \end{cases} \quad (10)$$

Given the initial state vector $\mathbf{y}_0 = [\mathbf{x}_0; t_i; t_f]$, the solver has to find the optimal state $\mathbf{y}$ that minimizes the objective function f_{obj} , defined as the total cost of the two-impulse transfer in terms of $\Delta v = v_{s/c} - v_{circ}$:

$$f_{obj}(\mathbf{y}) = \Delta v(\mathbf{x}_0, t_i, t_f) = \Delta v_i + \Delta v_f \quad (11)$$

where

$$\begin{cases} \Delta v_i &= \sqrt{(v_{x_0} - y_0)^2 + (v_{y_0} + x_0 + \mu)^2} - \sqrt{\frac{1-\mu}{r_i}} \\ \Delta v_f &= \sqrt{(v_{x_f} - y_f)^2 + (v_{y_f} + x_f + \mu - 1)^2} - \sqrt{\frac{\mu}{r_f}} \end{cases} \quad (12)$$

The final state $\mathbf{x}_f = [x_f; y_f; v_{x_f}; v_{y_f}]$ is retrieved from the propagation of the initial state $\mathbf{x}_0$ up to the final time t_f through the flow $\mathbf{x}_f = \varphi(\mathbf{x}_0, t_i; t_f)$.

Regarding the constraints, the linear inequality constraint is imposed for the condition $t_i \leq t_f$:

$$A = [0 \quad 0 \quad 0 \quad 0 \quad 1 \quad -1] \quad \text{and} \quad \mathbf{b} = \mathbf{0} \quad (13)$$

The nonlinear constraints, instead, are imposed just on the initial and final state of the transfer arc: $\mathbf{x}_0$ must lie at a distance r_i from Earth, while $\mathbf{x}_f$ must be at a distance r_f from the Moon. Furthermore, their velocity vector must be aligned with the circular velocity around Earth or Moon. These conditions translate into the following non-linear equality constraints $\mathbf{c}$.

$$\mathbf{c}(\mathbf{y}) = \begin{bmatrix} (x_0 + \mu)^2 + y_0^2 - r_i^2 \\ (x_0 + \mu)(v_{x_0} - y_0) + y_0(v_{y_0} + x_0 + \mu) \\ (x_f + \mu - 1)^2 + y_f^2 - r_f^2 \\ (x_f + \mu - 1)(v_{x_f} - y_f) + y_f(v_{y_f} + x_f + \mu - 1) \end{bmatrix} = \mathbf{0} \quad (14)$$

Finally, to guide the optimizer, it is useful to define the bounds box within the solver has to search. The lower and upper bounds $\mathbf{y}^{LB}$ and $\mathbf{y}^{UB}$ are defined on the departure and arrival circular orbit as follows.

1. From $\alpha \in [0, 2\pi]$, the initial position is constrained around the Earth, i.e. $x_0 \in [-\mu - r_i, -\mu + r_i]$ and $y_0 \in [-r_i, r_i]$.
2. From $\beta \in [1, \sqrt{2}]$, the initial velocity components are bounded by the Earth escape velocity $v_{esc} = \sqrt{2}v_{circ} = \sqrt{2\frac{1-\mu}{r_i}}$, i.e. $v_{x_0} \in [-v_{esc}, v_{esc}]$ and $v_{y_0} \in [-v_{esc}, v_{esc}]$.
3. From $\theta_i \in [0, 2\pi]$ and being $\theta_i = \omega_s t_i$, the initial time is constrained to ensure all possible configuration between the Earth–Moon system and the Sun, i.e. $t_i \in [0, -\frac{2\pi}{\omega_s}]$, where the minus is necessary for the relative rotation.
4. From $\delta \in [0, 23]$ (equivalent to $\approx [0, 100]$ days), the final time is limited by the maximum duration δ , i.e. $t_f \in [0, -\frac{2\pi}{\omega_s} + 23]$.

The optimization problem described above is then solved using the Matlab function `fmincon`. The propagations are performed using the Matlab solver `ode113` with absolute and relative tolerances set to $2.5 \cdot 10^{-14}$. The choice of such low tolerances was driven by the high non-linearity of the problem. As the optimization problem, the algorithm `active-set` is selected with a constraint tolerance of 10^{-10} , to ensure a small order of magnitude of the position error (order of meters).

2.2.1 Without derivatives

Initially, the optimization is run without providing any derivative. Thus, the analytical gradients of $f_{obj}(\mathbf{y})$ and $\mathbf{c}(\mathbf{y})$ are not supplied to the solver. Furthermore, all integrations necessary to build $f_{obj}(\mathbf{y})$ and $\mathbf{c}(\mathbf{y})$ and to propagate the resulting orbit do not include the STM, as it is the first-order Taylor expansion of the flow ($\Phi(\mathbf{x}_0, t_i; t) = \frac{\partial \varphi}{\partial \mathbf{x}_0}(\mathbf{x}_0, t_i; t)$). Figure 5 shows the resulting transfer in both the Earth–Moon rotating and Earth-centered inertial adimensional reference frames. It can be observed that the transfer leg is now able to correctly reach the Moon, and that the insertion is tangent to the orbit, as visible from the Arrival Zoom.

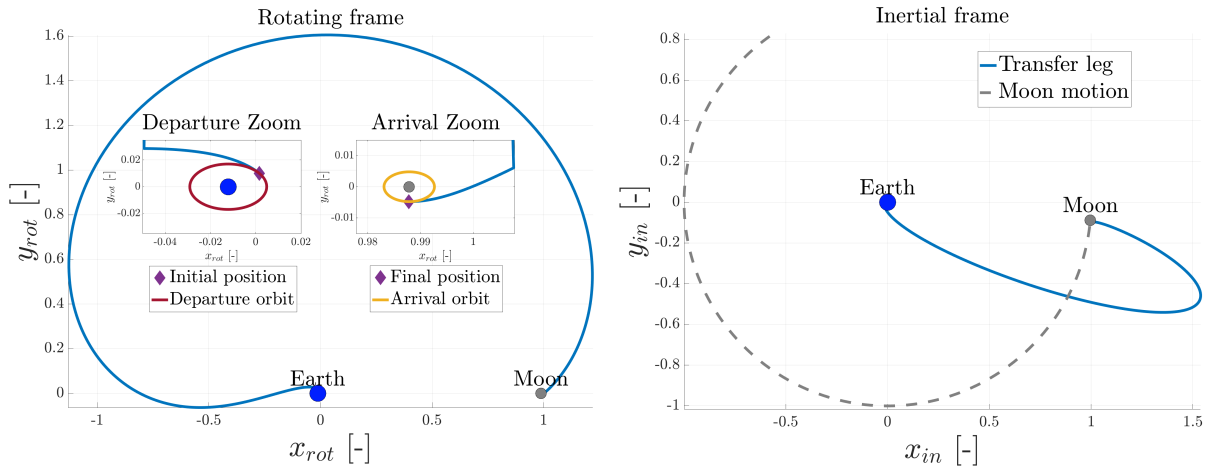


Figure 5: Simple shooting solution without derivatives in both adimensional frames

After **43** iterations and **365** function evaluations, the solver finds a transfer cost of $\Delta v = 4.006839[-] = 4.105167 \text{ km/s}$, with the worst equality constraint of $7.316691 \cdot 10^{-11}$.

2.2.2 With derivatives

Then, the gradients of $f_{obj}(\mathbf{y})$ and $\mathbf{c}(\mathbf{y})$ are provided to the optimization in analytical form to improve the convergence of the algorithm. As a consequence, all integrations now include the STM.

The gradient of $f_{obj}(\mathbf{y})$ is expressed as:

$$\nabla f_{obj} = \begin{bmatrix} \frac{\partial f_{obj}}{\partial x_0} \\ \frac{\partial f_{obj}}{\partial t_i} \\ \frac{\partial f_{obj}}{\partial t_f} \end{bmatrix} = \begin{bmatrix} \frac{\partial \Delta v_i}{\partial x_0} + \Phi^T(\mathbf{x}_0, t_i; t_f) \frac{\partial \Delta v_f}{\partial x_f} \\ - \left(\frac{\partial \Delta v_f}{\partial x_f} \right)^T \Phi(\mathbf{x}_0, t_i; t_f) \mathbf{f}(\mathbf{x}_0, t_i) \\ \left(\frac{\partial \Delta v_f}{\partial x_f} \right)^T \mathbf{f}(\mathbf{x}_f, t_f) \end{bmatrix} \quad (15)$$

where:

$$\frac{\partial \Delta v_i}{\partial x_0} = \frac{1}{\sqrt{(\dot{x}_0 - y_0)^2 + (\dot{y}_0 + x_0 + \mu)^2}} \begin{bmatrix} \dot{y}_0 + x_0 + \mu \\ -(\dot{x}_0 - y_0) \\ \dot{x}_0 - y_0 \\ \dot{y}_0 + x_0 + \mu \end{bmatrix} \quad (16)$$

$$\frac{\partial \Delta v_f}{\partial x_f} = \frac{1}{\sqrt{(\dot{x}_f - y_f)^2 + (\dot{y}_f + x_f + \mu - 1)^2}} \begin{bmatrix} \dot{y}_f + x_f + \mu - 1 \\ -(\dot{x}_f - y_f) \\ \dot{x}_f - y_f \\ \dot{y}_f + x_f + \mu + 1 \end{bmatrix} \quad (17)$$

and $\mathbf{f}(\mathbf{x}, t)$ representing the RHS of the PBRFBP (Appendix C).

As shown, the computation requires the STM. Consequently, along with the propagation of dynamics, the initial value problem must also be solved. Here, $A(t)$ is the Jacobian of the dynamics, and its expression is reported in Appendix E.

$$\begin{cases} \dot{\Phi}(t_i; t) = A(t) \Phi(t_i; t), \\ \Phi(t_i; t_i) = \mathbf{I}_{4 \times 4}. \end{cases} \quad (18)$$

The gradient of $\mathbf{c}(\mathbf{y})$ is constructed as:

$$\nabla \mathbf{c} = [\nabla \mathbf{c}_1 \nabla \mathbf{c}_2 \nabla \mathbf{c}_3 \nabla \mathbf{c}_4] \quad (19)$$

where the columns are defined as:

$$\nabla \mathbf{c}_1 = [2(x_0 + \mu) \quad 2y_0 \quad 0 \quad 0 \quad 0 \quad 0]^T \quad (20)$$

$$\nabla \mathbf{c}_2 = [\dot{x}_0 \quad \dot{y}_0 \quad x_0 + \mu \quad y_0 \quad 0 \quad 0]^T \quad (21)$$

$$\nabla \mathbf{c}_3 = \left[\Phi^T(t_i; t_f) \frac{\partial \mathbf{c}_3}{\partial \mathbf{x}_f} - \frac{\partial \mathbf{c}_3}{\partial \mathbf{x}_f} \Phi^T(t_i; t_f) \mathbf{f}(\mathbf{x}_0, t_i) \left(\frac{\partial \mathbf{c}_3}{\partial \mathbf{x}_f} \right)^T \mathbf{f}(\mathbf{x}_f, t_f) \right]^T \quad (22)$$

$$\nabla \mathbf{c}_4 = \left[\Phi^T(t_i; t_f) \frac{\partial \mathbf{c}_4}{\partial \mathbf{x}_f} - \frac{\partial \mathbf{c}_4}{\partial \mathbf{x}_f} \Phi^T(t_i; t_f) \mathbf{f}(\mathbf{x}_0, t_i) \left(\frac{\partial \mathbf{c}_4}{\partial \mathbf{x}_f} \right)^T \mathbf{f}(\mathbf{x}_f, t_f) \right]^T \quad (23)$$

where:

$$\frac{\partial \mathbf{c}_3}{\partial \mathbf{x}_f} = [2(x_f + \mu - 1) \quad 2y_f \quad 0 \quad 0]^T \quad (24)$$

$$\frac{\partial \mathbf{c}_4}{\partial \mathbf{x}_f} = [\dot{x}_f \quad \dot{y}_f \quad x_f + \mu - 1 \quad y_f]^T \quad (25)$$

The resulting transfer in both the Earth–Moon rotating and Earth–centered inertial adimensional reference frames are shown in Figure 6, while the results are presented in Table 6

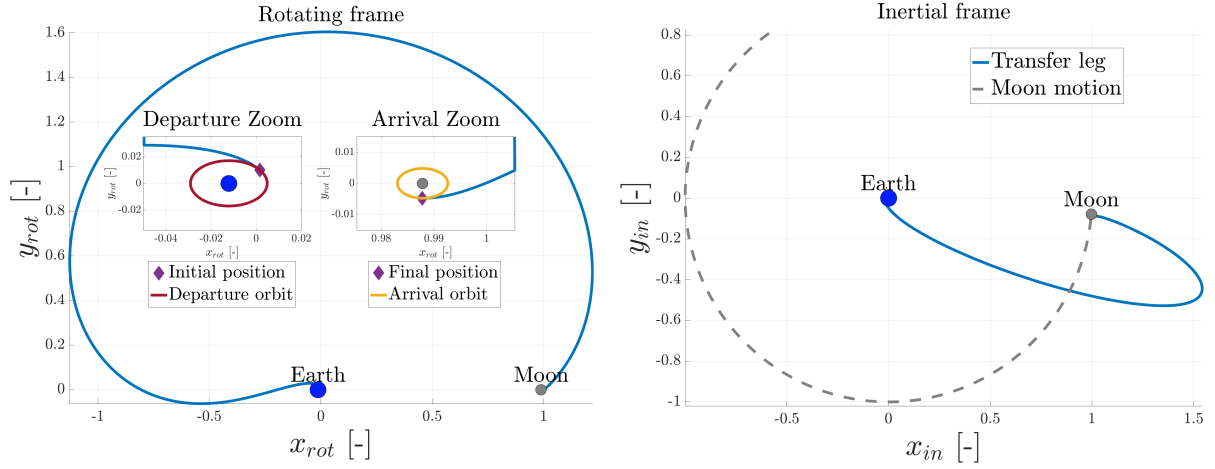


Figure 6: Simple shooting solution with derivatives in both adimensional frames

Gradients	$r_{x,0}$ [DU]	$r_{y,0}$ [DU]	$v_{x,0}$ [VU]	$v_{y,0}$ [VU]	t_i [TU]	t_f [TU]
False	+0.001611	+0.010026	−6.298521	+8.645788	2.189	6.195
True	+0.001621	+0.010013	−6.290520	+8.651573	2.199	6.204

Table 6: Simple shooting solutions in the Earth-Moon rotating frame.

After **14** iterations and **31** function evaluations, the solver finds a transfer cost of $\Delta v = 4.006831[-] = 4.105159 \text{ km/s}$, with the worst equality constraint of $6.022177 \cdot 10^{-15}$. As expected, providing the derivatives improves the overall convergence in terms of accuracy, computational cost, time and optimality.

2.3 Multiple shooting

Starting from the same initial guess, it is possible to solve the problem with multiple shooting taking $N = 4$ points of the transfer arc equally spaced in time, thus subdividing the arc into $N - 1 = 3$ smaller arcs associated with equal time intervals. Keeping the same NLP problem form, now the vector of variables $\mathbf{y}$ becomes $4N + 2$ long, so a column vector of 18 elements:

$$\mathbf{y} = [\mathbf{x}_1 \quad \mathbf{x}_2 \quad \mathbf{x}_3 \quad \mathbf{x}_4 \quad t_1 \quad t_N]^T \quad (26)$$

The objective function is then:

$$f_{obj}(\mathbf{y}) = \Delta v_1 + \Delta v_2 = \Delta v_i(\mathbf{x}_1) + \Delta v_f(\mathbf{x}_N) \quad (27)$$

The nonlinear equality constraints are expressed as:

$$\mathbf{c}(\mathbf{y}) = [\mathbf{z}_1 \quad \mathbf{z}_2 \quad \mathbf{z}_3 \quad \Psi_1 \quad \Psi_N] \quad (28)$$

where Ψ_1 and Ψ_N are equivalent to the constraints on the initial and final conditions in the simple shooting formulation. The defect functions $\mathbf{z}_i$ ensure continuity between the solution at the j -th node integrated over the j -th time interval and the discretized solution at the $(j + 1)$ -th node:

$$\mathbf{z}_i = \varphi(\mathbf{x}_j, t_j; t_{j+1}) - \mathbf{x}_{j+1} \quad (j = 1, \dots, N - 1) \quad (29)$$

In the multiple shooting formulation, both linear and nonlinear inequality constraints are necessary. The linear constraint ensures that $t_1 \leq t_N$, and is expressed by:

$$A\mathbf{y} \leq b, \text{ where} \quad (30)$$

$$A = [\mathbf{0}_{1 \times 16} \quad 1 \quad -1] \quad \text{and} \quad b = 0 \quad (31)$$

On the other hand, nonlinear inequality constraints prevent the optimization process from finding solutions that result in impacts with either the Earth or the Moon:

$$\mathbf{g}(\mathbf{y}) = [\mathbf{g}_1 \quad \mathbf{g}_2 \quad \mathbf{g}_3 \quad \mathbf{g}_4]^T < 0, \quad \text{with} \quad (32)$$

$$\mathbf{g}_j = \begin{bmatrix} \left(\frac{R_E}{DU}\right)^2 - (x_j + \mu)^2 - y_j^2 \\ \left(\frac{R_M}{DU}\right)^2 - (x_j + \mu - 1)^2 - y_j^2 \end{bmatrix}, \quad j = 1, \dots, N. \quad (33)$$

To improve the convergence properties of the algorithm, the gradients of $f_{obj}(\mathbf{y})$, $\mathbf{c}(\mathbf{y})$ and $\mathbf{g}(\mathbf{y})$ are provided to `fmincon`. First, for the objective function, the gradient can be expressed as:

$$\nabla f_{obj}(\mathbf{y}) = \begin{bmatrix} \frac{\partial \Delta v_1^T}{\partial \mathbf{x}_1} & \mathbf{0}_{1 \times 8} & \frac{\partial \Delta v_2^T}{\partial \mathbf{x}_N} & 0 & 0 \end{bmatrix}^T \quad (34)$$

Second, the gradient of the nonlinear equality constraints can be written as:

$$\nabla \mathbf{c} = \begin{bmatrix} \Phi(t_1, t_2) & -\mathbf{I}_{4 \times 4} & \mathbf{0}_{4 \times 4} & \mathbf{0}_{4 \times 4} & \frac{\partial \mathbf{z}_1}{\partial t_1} & \frac{\partial \mathbf{z}_1}{\partial t_N} \\ \mathbf{0}_{4 \times 4} & \Phi(t_2, t_3) & -\mathbf{I}_{4 \times 4} & \mathbf{0}_{4 \times 4} & \frac{\partial \mathbf{z}_2}{\partial t_1} & \frac{\partial \mathbf{z}_2}{\partial t_N} \\ \mathbf{0}_{4 \times 4} & \mathbf{0}_{4 \times 4} & \Phi(t_3, t_N) & -\mathbf{I}_{4 \times 4} & \frac{\partial \mathbf{z}_3}{\partial t_1} & \frac{\partial \mathbf{z}_3}{\partial t_N} \\ \frac{\partial \Psi_1}{\partial \mathbf{x}_1} & \mathbf{0}_{2 \times 4} & \mathbf{0}_{2 \times 4} & \mathbf{0}_{2 \times 4} & \mathbf{0}_{2 \times 1} & \mathbf{0}_{2 \times 1} \\ \mathbf{0}_{2 \times 4} & \mathbf{0}_{2 \times 4} & \mathbf{0}_{2 \times 4} & \frac{\partial \Psi_N}{\partial \mathbf{x}_N} & \mathbf{0}_{2 \times 1} & \mathbf{0}_{2 \times 1} \end{bmatrix} \quad (35)$$

where:

$$\begin{aligned} \frac{\partial \mathbf{z}_j}{\partial t_1} &= -\frac{N-j}{N-1} \Phi(t_j, t_{j+1}) \mathbf{f}(t_j, \mathbf{x}_j) + \frac{N-j-1}{N-1} \mathbf{f}(t_j, \varphi(\mathbf{x}_j, t_j, t_{j+1})), & j, \dots, N-1 \\ \frac{\partial \mathbf{z}_j}{\partial t_N} &= -\frac{j-1}{N-1} \Phi(t_j, t_{j+1}) \mathbf{f}(t_j, \mathbf{x}_j) + \frac{j}{N-1} \mathbf{f}(t_j, \varphi(\mathbf{x}_j, t_j, t_{j+1})), & j, \dots, N-1 \\ \frac{\partial \Psi_1}{\partial \mathbf{x}_1} &= \begin{bmatrix} 2(x_1 + \mu) & 2y_1 & 0 & 0 \\ \dot{x}_1 & \dot{y}_1 & x_1 + \mu & y_1 \end{bmatrix} \\ \frac{\partial \Psi_N}{\partial \mathbf{x}_N} &= \begin{bmatrix} 2(x_N + \mu - 1) & 2y_N & 0 & 0 \\ \dot{x}_N & \dot{y}_N & x_N + \mu - 1 & y_N \end{bmatrix}. \end{aligned} \quad (36)$$

Finally, the gradient of the nonlinear inequality constraints is retrieved as follows:

$$\nabla \mathbf{g} = \begin{bmatrix} \frac{\partial \mathbf{g}_1}{\partial \mathbf{x}_1} & \mathbf{0}_{2 \times 4} & \mathbf{0}_{2 \times 4} & \mathbf{0}_{2 \times 4} & 0 & 0 \\ \mathbf{0}_{2 \times 4} & \frac{\partial \mathbf{g}_2}{\partial \mathbf{x}_2} & \mathbf{0}_{2 \times 4} & \mathbf{0}_{2 \times 4} & 0 & 0 \\ \mathbf{0}_{2 \times 4} & \mathbf{0}_{2 \times 4} & \frac{\partial \mathbf{g}_3}{\partial \mathbf{x}_3} & \mathbf{0}_{2 \times 4} & 0 & 0 \\ \mathbf{0}_{2 \times 4} & \mathbf{0}_{2 \times 4} & \mathbf{0}_{2 \times 4} & \frac{\partial \mathbf{g}_4}{\partial \mathbf{x}_4} & 0 & 0 \end{bmatrix} \quad (37)$$

where:

$$\frac{\partial \mathbf{g}_i}{\partial \mathbf{x}_i} = \begin{bmatrix} -2(x_j + \mu) & -2y_j & 0 & 0 \\ -2(x_j + \mu - 1) & -2y_j & 0 & 0 \end{bmatrix} \quad j, \dots, N-1 \quad (38)$$

where the last two columns are null because there is no constraint depending on time.

Thus, the NLP problem related to the multiple shooting becomes:

$$\begin{aligned} \min_{\mathbf{y}} \quad & f_{obj}(\mathbf{y}) \\ \text{s.t.} \quad & \begin{cases} A\mathbf{y} \leq b, \\ \mathbf{c}(\mathbf{y}) = 0, \\ \mathbf{g}(\mathbf{y}) < 0. \end{cases} \end{aligned} \quad (39)$$

The initial guess for the multiple shooting method is a crucial point for the optimization process. Through a uniform time grid, the initial guess $\mathbf{x}_0$ from Section 2.1 is integrated using `ode113`, enforcing the time grid to obtain the state at $N = 4$ equally spaced time instants. The multiple shooting initial guess is formed as shown in Equation 26, where t_1 and t_N are the initial and final time guesses from the simple shooting method.

The resulting orbit is exhibited in both the Earth–Moon rotating and Earth-centered inertial adimensional reference frames in Figure 7, while Table 7 reports the results.

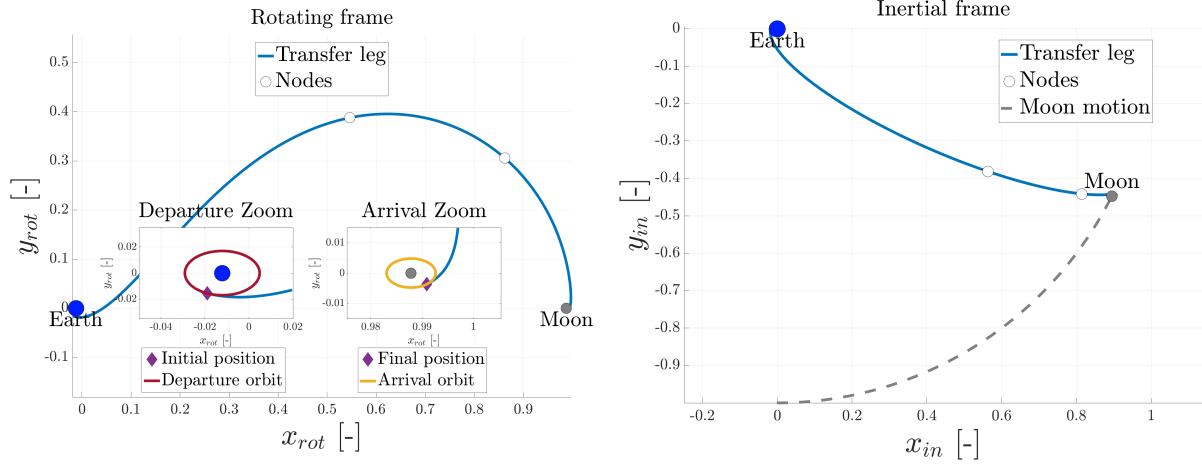


Figure 7: Multiple shooting solution in both adimensional frames

$r_{x,0}$ [DU]	$r_{y,0}$ [DU]	$v_{x,0}$ [VU]	$v_{y,0}$ [VU]	t_i [TU]	t_f [TU]
− 0.018926	−0.015621	+9.785206	−4.244065	4.711	5.819

Table 7: Multiple shooting solution in the Earth-Moon rotating frame.

This solution is obtained with the same constraint tolerance used for the simple shooting, but settings `MaxIterations` = $1 \cdot 10^4$ and `MaxFunctionEvaluations` = $4 \cdot 10^4$ are necessary to avoid the solver stopping before finding the solution (an intermediate feasible solution found without these settings is reported in Appendix F). This leads to prolonged run times for the code: the solution is found in **6703** iterations and **30599** function evaluations, with a maximum constraint error of $\mathbf{2.523093} \cdot 10^{-12}$. The benefit of this long run is the final transfer cost resulting in $\Delta v = \mathbf{3.860313}[-] = \mathbf{3.955045 km/s}$. The transfer time $t_f - t_i$ can be translated into days through TU, resulting in $\Delta t \approx 4.812 days$.

These results and the transfer shape obtained are consistent with what is presented in the reference article [1], here reported in Figure 8. The transfer is close to the set of transfer belonging to the Pareto front (black lines), identified in the reference article [1]. This indicates that the transfer achieved with the multiple shooting method strikes a good balance between Δv and Δt . In fact, this solution is very different with respect to the simple shooting one in terms of optimal initial states, transfer time and cost. The latter are found to be lower in the multiple shooting case because a Pareto-optimal solution has been found, while the one from

simple shooting still belongs to one of the curves but it is not close to one of the red optimal points.

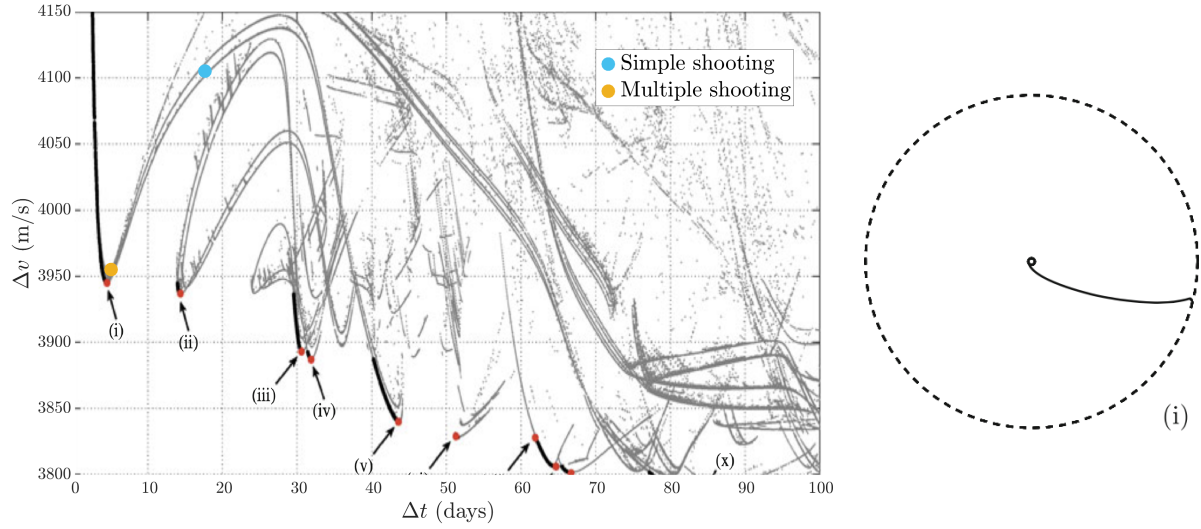


Figure 8: Pareto front solution and Transfer shape associated to a very short Δt [1]

Finally, Table 8 shows the comparison between the three approaches: simple shooting with or without derivatives and multiple shooting. As expected, the multiple shooting method is computationally more expensive than the simple shooting one. Nevertheless, the solution is notably improved in terms of both Δv and Δt . Furthermore, it can be noted that the comparison between Figure 6 and Figure 7 shows that the trajectory obtained with the multiple shooting method is more direct, resembling the shape of a Hohmann transfer.

Method	Iter	FuncEval	MaxConstr	$\Delta v[-]$	$\Delta v[km/s]$
Simple without gradients	43	365	$7.316691 \cdot 10^{-11}$	4.006839	4.105167
Simple with gradients	14	31	$6.022177 \cdot 10^{-15}$	4.006831	4.105159
Multiple shooting	6703	30599	$2.523093 \cdot 10^{-12}$	3.860313	3.955045

Table 8: Comparison between the two approaches

2.4 N-body propagation

The propagation of the initial condition obtained in Section 2.2 using the PBRFBP is compared to an n-body propagation, which accounts for a more complete gravitational model. Since the n-body propagation is not a planar problem, the initial conditions are adjusted to include $z(t_i) = 0km$ and $v_z(t_i) = 0km/s$.

To perform the propagation, it is necessary to determine the initial epoch t_i , identifying when the relative positions of the Earth, Moon, and Sun matches those in Section 2.2. This is achieved by solving a root-finding problem where the function to be zeroed is $f(t) = \theta(t) - \theta_i$. The angle θ_i corresponding to the initial time t_i identified in Section 2.2 can be computed as:

$$\theta_{\text{target}} = \omega_s t_{\text{target}} \approx 4.2488 \text{ rad} \approx 243.4369 \text{ deg} \quad (40)$$

The initial epoch is searched between 2024-09-28 and 2024-10-25. This time span was chosen because the Moon's orbital period around the Earth is approximately 27 days, allowing θ to cover all range $[0, 2\pi]$.

Through the function `cspice_spkezr`, the full Moon state for each instant of the time span can be retrieved, expressed in the frame ECLIPJ2000 centered at the EMB (Earth-Moon Barycen-

ter). In particular, the Moon state in this inertial frame $\mathbf{x}_M(t) = [\mathbf{r}_{M,in}(t), \mathbf{v}_{M,in}(t)]$ can be used to build the rotation matrix $\mathbf{R}(t)$ to pass from the rotating frame to the inertial one. The columns of this matrix are the unit vectors $[\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}]$ of the rotating frame expressed in the inertial frame:

$$\begin{aligned}\hat{\mathbf{r}} &= \frac{\mathbf{r}_{M,in}(t)}{r_{M,in}(t)}, \hat{\mathbf{v}} = \frac{\mathbf{v}_{M,in}(t)}{v_{M,in}(t)}, \hat{\mathbf{h}} = \frac{\hat{\mathbf{r}} \times \hat{\mathbf{v}}}{\|\hat{\mathbf{r}} \times \hat{\mathbf{v}}\|}, \\ \hat{\mathbf{i}} &= \hat{\mathbf{r}}, \\ \hat{\mathbf{j}} &= \hat{\mathbf{h}} \times \hat{\mathbf{r}}, \\ \hat{\mathbf{k}} &= \hat{\mathbf{h}}, \\ \mathbf{R}(t) &= [\hat{\mathbf{i}} \quad \hat{\mathbf{j}} \quad \hat{\mathbf{k}}]\end{aligned}\tag{41}$$

It can be noticed that $\hat{\mathbf{j}} \neq \hat{\mathbf{v}}$ as these two unit vectors are equal just at the apsidal points. It would be absolutely true if the Moon orbit was perfectly circular, but using the SPK kernels the true Moon orbit is known to be not circular. The matrix is computed at each time step of the considered time span and can be used to find the time evolution of the angle θ . The position vector of the Sun $\mathbf{r}_{S,in}(t)$ is computed at each time instant. $\theta(t)$ is the angle between the projection of the Sun's position relative to the EMB onto the Earth-Moon plane and the EMB-Moon axis, so the time evolution of the angle can be found:

$$\mathbf{r}_{S,projected}(t) = \mathbf{R}^T(t) \mathbf{r}_{S,in}(t) = \{x_{S,proj}(t) \ y_{S,proj}(t) \ z_{S,proj}(t)\}\tag{42}$$

$$\theta(t) = \arctan\left(\frac{y_{S,proj}(t)}{x_{S,proj}(t)}\right)\tag{43}$$

The described zero-finding problem was solved using Matlab function `fsolve`, with an `OptimalityTolerance` of 10^{-13} . The initial guess was selected graphically based on the angle evolution shown in Figure 9, by plotting $\theta(t) \in [0, 2\pi]$ in time and the initial angle θ_i .

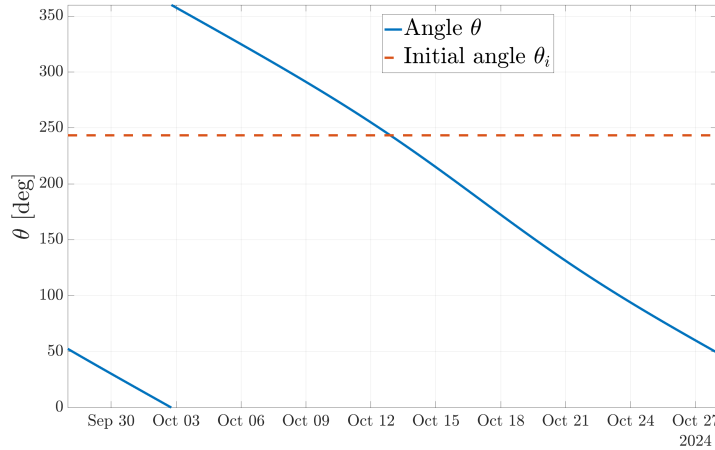


Figure 9: θ time evolution

The ephemeris time in which the time evolution $\theta(t)$ crosses the threshold set by θ_i is $\approx 7.820 \cdot 10^8 s$. Then, the final epoch is obtained by adding the transfer time calculated in Section 2.2 ($\Delta t = t_f - t_i \approx 17.392 days$) to the initial epoch just found.

The results, including t_i , t_f , and $\mathbf{x}_0$, are reported in Table 9.

Symbol	Calendar epoch (UTC)		
t_i	2024-10-12T21:29:54.659		
t_f	2024-10-30T06:54:24.896		
$r_{x,0}$ [km]	$r_{y,0}$ [km]	$r_{z,0}$ [km]	
−6225.08387423	+2021.66858595	0.0	
$v_{x,0}$ [km/s]	$v_{y,0}$ [km/s]	$v_{z,0}$ [km/s]	
−3.39049128	−10.43993697	0.0	

Table 9: Initial epoch, final epoch, and initial state in Earth-centered inertial frame.

Now, it is possible to propagate the initial state within this time interval using a n-body propagator which accounts for the perturbing gravitational accelerations of other bodies with an integration center about the Earth, considering as perturbing bodies the Sun, Mercury, Venus, Mars, Jupiter, Saturn, Uranus, Neptune, Pluto and the Moon. The trajectories, in the Earth-centered ECLIPJ2000 reference frame, obtained with the PBRFBP (transformed into dimensional units through DU) and the n-body propagation are shown in Figure 10. It can be observed that, initially, the two trajectories almost overlap. However, over time, evident out-of-plane components of the transfer arc in the n-body case remark the difference in terms of equations of motion and perturbing terms. It is important to note that in the 3D view, the scale of the z_{in} axis is two orders of magnitude smaller than that of the x_{in} and y_{in} axes, in order to appreciate the non-planarity of the n-body propagation.

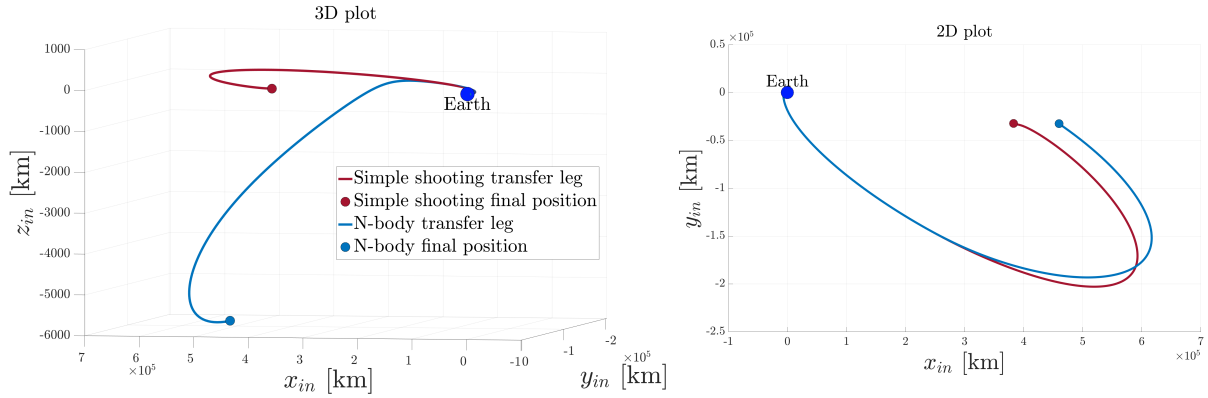


Figure 10: 3D and 2D plot of the comparison in Earth-centered ECLIPJ2000 reference frame

3 Continuous guidance

Exercise 3

A low-thrust option is being considered to perform an orbit raising maneuver using a low-thrust propulsion system in Earth orbit. The spacecraft is released on a circular orbit on the equatorial plane at an altitude of 800 km and has to reach an orbit inclined by 0.75 deg on the equatorial plane at 1000 km. This orbital regime is characterized by a large population of resident space objects and debris, whose spatial density q can be expressed as:

$$q(\rho) = \frac{k_1}{k_2 + \left(\frac{\rho - \rho_0}{DU}\right)^2}$$

where ρ is the distance from the Earth center. The objective is to design an optimal orbit raising that minimizes the risk of impact, that is to minimize the following objective function

$$F(t) = \int_{t_0}^{t_f} q(\rho(t)) dt.$$

The parameters and reference Distance Unit to be considered are provided in Table 10.

Symbol	Value	Units	Meaning
h_i	800	km	Altitude of departure orbit
h_f	1000	km	Altitude of arrival orbit
Δi	0.75	deg	Inclination change
R_e	6378.1366	km	Earth radius
μ	398600.435	km ³ /s ²	Earth gravitational parameter
ρ_0	$750 + R_e$	km	Reference radius for debris flux
k_1	1×10^{-5}	DU ⁻¹	Debris spatial density constant 1
k_2	1×10^{-4}	DU ²	Debris spatial density constant 2
m_0	1000	kg	Initial mass
$T_{\max}$	3.000	N	Maximum thrust
I_{sp}	3120	s	Specific impulse
DU	7178.1366	km	Distance Unit
MU	m_0	kg	Mass Unit

Table 10: Problem parameters and constants. The units of time TU and velocity VU can be computed imposing that the scaled gravitational parameter $\bar{\mu} = 1$.

- 1) Plot the debris spatial density $q(\rho) \in [h_i - 100; h_f + 100]$ km and compute the initial state and target orbital state, knowing that: i) the initial and final state are located on the x -axis of the equatorial J2000 reference frame; ii) the rotation of the angle Δi is performed around the x -axis of the equatorial J2000 reference frame (RAAN = 0).
- 2) Adimensionalize the problem using as reference length $DU = \rho_i = h_i + R_e$ and reference mass $MU = m_0$, imposing that $\mu = 1$. Report all the adimensionalized parameters.
- 3) Using the PMP, write down the spacecraft equations of motion, the costate dynamics, and the zero-finding problem for the unknowns $\{\lambda_0, t_f\}$ with the appropriate transversality condition. **Hint:** the spacecraft has to reach the target state computed in point 1).
- 4) Solve the problem considering the data provided in Table 10. To obtain an initial guess for the costate, generate random numbers such that $\lambda_{0,i} \in [-250; +250]$ while $t_f \approx 20\pi$.

Report the obtained solution in terms of $\{\lambda_0, t_f\}$ and the error with respect to the target. Assess your results exploiting the properties of the Hamiltonian in problems that are not time-dependent and time-optimal solution. Plot the evolution of the components of the primer vector α in a NTW reference frame[†].

- 5) Solve the problem for a lower thrust level $T_{\max} = 2.860$ N. Compare the new solution with the one obtained in the previous point. **Hint:** exploit numerical continuation.

(11 points)

3.1 Debris density, Initial and Final states

The debris spatial density is function of the orbital radius module ρ , and it is depicted in Figure 11 in the interval $h \in [h_i - 100, h_f + 100] = [700, 1100]$.

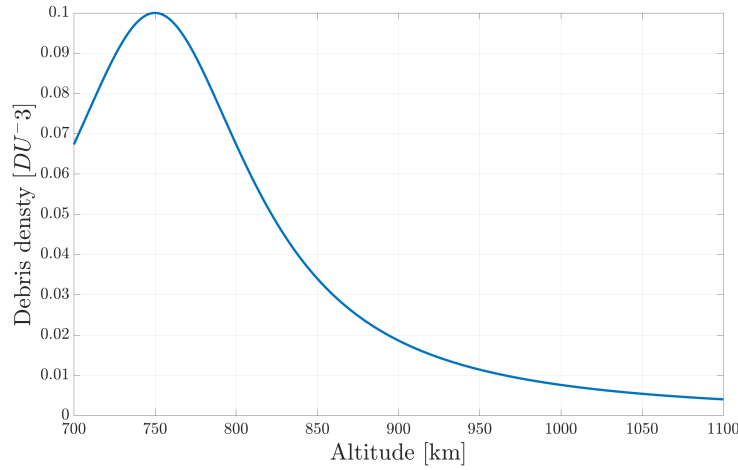


Figure 11: Debris spatial density

Knowing that the initial orbit is along the equator, the states are expressed in the Earth-centered equatorial J2000 inertial frame and the initial position vector is located on the x -axis. The initial x -component is equal to the initial orbital radius and the circular velocity is aligned with the y -axis:

$$r_{x,i} = \rho_i \quad , \quad v_{y,i} = \sqrt{\frac{\mu}{\rho_i}} \quad (44)$$

Regarding the target position, a similar reasoning can be done for $r_{x,f}$. While, knowing that there is a difference in inclination dictated by a clockwise rotation Δi around the x -axis, the target orbit circular velocity $v_{circ,f}$ is still perpendicular to the position vector but it shall be decomposed into y and z components:

$$v_{y,f} = \sqrt{\frac{\mu}{\rho_f}} \cos \Delta i \quad , \quad v_{z,f} = \sqrt{\frac{\mu}{\rho_f}} \sin \Delta i \quad (45)$$

[†]The T-axis is aligned with the velocity, the N-axis is normal to the angular momentum, while the W-axis is pointing inwards, i.e., towards the Earth.

The two retrieved states are reported in Table 11.

$r_{x,i} [km]$	$r_{y,i} [km]$	$r_{z,i} [km]$	$v_{x,i} [km/s]$	$v_{y,i} [km/s]$	$v_{z,i} [km/s]$
+7178.136600	0000.000000	0000.000000	0.00000000	+7.45183148	0.00000000
$r_{x,f} [km]$	$r_{y,f} [km]$	$r_{z,f} [km]$	$v_{x,f} [km/s]$	$v_{y,f} [km/s]$	$v_{z,f} [km/s]$
+7378.136600	0000.000000	0000.000000	0.00000000	+7.34950906	0.09621034

Table 11: Initial and target state in Earth-centered equatorial J2000 inertial frame.

3.2 Adimensionalized problem

The first step is to find the time unit TU and the velocity unit VU which allow to adimensionalize the entire problem. The velocity unit is found through the adimensional equation of the circular velocity with the adimensional Earth parameter $\bar{\mu} = 1$. The time unit is found simply by inverting the definition of the velocity.

$$\frac{v_{\text{circ}}}{VU} = \sqrt{\frac{1}{\rho/DU}} \Rightarrow VU = \frac{v_{\text{circ},i}}{\sqrt{DU/\rho_i}} \Rightarrow TU = \frac{DU}{VU} \quad (46)$$

The values of these reference units are reported in Table 12, along with all adimensionalized parameters. The resulting adimensionalized states through DU and VU are reported in Table 13. Note that gravity is considered constant and not a function of the Earth radius and Earth gravitational parameter.

Symbol	Value
$\bar{h}_i$	0.111450
$\bar{h}_f$	0.139312
$\bar{R}_e$	0.888550
$\bar{\rho}_i$	1
$\bar{\rho}_f$	1.027862
$\bar{v}_{\text{circ},i}$	1
$\bar{v}_{\text{circ},f}$	0.986353
$\bar{\rho}_0$	0.993034
$\bar{m}_0$	1
$\bar{T}_{\text{max}}$	0.0003878
$\bar{I}_{\text{sp}}$	3.2389624
$\bar{g}_0$	1.267670
DU	7178.1366 km
MU	1000 kg
VU	7.451831 km/s
TU	963.271462 s

Table 12: Parameters in adimensional units and reference units

$r_{x,i} [km]$	$r_{y,i} [km]$	$r_{z,i} [km]$	$v_{x,i} [km/s]$	$v_{y,i} [km/s]$	$v_{z,i} [km/s]$
+1.000000	0000.000000	0000.000000	0.00000000	+1.00000000	0.00000000
$r_{x,f} [km]$	$r_{y,f} [km]$	$r_{z,f} [km]$	$v_{x,f} [km/s]$	$v_{y,f} [km/s]$	$v_{z,f} [km/s]$
+1.027862	0000.000000	0000.000000	0.00000000	+0.98626882	0.01291097

Table 13: Adimensionalized states in Earth-centered equatorial J2000 inertial frame

3.3 PMP problem

Now, the Pontryagin Maximum Principle (PMP) method is applied and the optimal control problem (OCP) can be stated as follows:

$$\min_{(\hat{\alpha}, u) \in \Omega} J(\mathbf{x}, \mathbf{u}, t) = \int_{t_0}^{t_f} q(\rho(t)) dt \quad \text{s.t.} \quad \begin{cases} \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}, t) \\ \mathbf{x}(t_0) = \mathbf{x}_0 \\ \mathbf{r}(t_f) = \mathbf{r}_f \\ \mathbf{v}(t_f) = \mathbf{v}_f \\ \lambda_m(t_f) = 0 \end{cases} \quad (47)$$

Here, $q(\rho(t))$ is the debris density function, $\mathbf{f}$ is the right-hand side of the dynamics formulated below, and Ω is the set of admissible control actions: $\Omega = (\hat{\alpha}, u) : \hat{\alpha} = 1, u \in [0, 1]$. The condition $\lambda_m(t_f) = 0$ is present since $m(t_f)$ cannot be determined a-priori.

$$\mathbf{f}(\mathbf{x}, \mathbf{u}, t) = \dot{\mathbf{x}} = \begin{bmatrix} \dot{\bar{\mathbf{r}}} \\ \dot{\bar{\mathbf{v}}} \\ \dot{\bar{m}} \end{bmatrix} = \begin{bmatrix} \bar{\mathbf{v}} \\ -\frac{\bar{\mu}}{\bar{r}^3} \bar{\mathbf{r}} + u \frac{\bar{T}_{\max}}{\bar{m}} \hat{\alpha} \\ -u \frac{\bar{T}_{\max}}{\bar{I}_{sp} \bar{g}_0} \end{bmatrix} \quad (48)$$

The Hamiltonian is defined as $H = q(\rho) + \boldsymbol{\lambda} \cdot \mathbf{f}$, with the explicit expression given by:

$$H = q(\rho) + \boldsymbol{\lambda}_r \cdot \bar{\mathbf{v}} - \frac{\bar{\mu}}{\bar{r}^3} \bar{\mathbf{r}} \cdot \boldsymbol{\lambda}_v + u \frac{\bar{T}_{\max}}{\bar{m}} \boldsymbol{\lambda}_v \cdot \hat{\alpha} - \lambda_m u \frac{\bar{T}_{\max}}{\bar{I}_{sp} \bar{g}_0} \quad (49)$$

The PMP states that the admissible control action $\mathbf{u} \in \Omega$ that minimizes the functional J is the one that minimizes the Hamiltonian:

$$\mathbf{u}^* = (\hat{\alpha}^*, u^*) := \arg \min_{(\hat{\alpha}^*, u^*) \in \Omega} H(\mathbf{x}, \boldsymbol{\lambda}, \mathbf{u}) \quad (50)$$

Looking at the Hamiltonian expression, it is possible to note that the thrust direction that minimizes the Hamiltonian is the Primer Vector, which is defined as:

$$\hat{\alpha}^* = -\frac{\boldsymbol{\lambda}_v}{\lambda_v} \Rightarrow H = q(\rho) + \boldsymbol{\lambda}_r \cdot \bar{\mathbf{v}} - \frac{\bar{\mu}}{\bar{r}^3} \bar{\mathbf{r}} \cdot \boldsymbol{\lambda}_v + \frac{\bar{T}_{\max}}{\bar{I}_{sp} \bar{g}_0} u S_t \quad (51)$$

$$\text{where } S_t = -\frac{\lambda_v}{\bar{m}} \bar{I}_{sp} \bar{g}_0 - \lambda_m \quad (52)$$

is the switching function. The sign of S_t determines the control policy and, consequently, the time history of the throttling factor u . The factor $\frac{\lambda_v}{\bar{m}} \bar{I}_{sp} \bar{g}_0$ is positive, as it consists of positive quantities. Furthermore, λ_m is always positive, since it must be zero at t_f and, since $u \in [0, 1]$, its time derivative $\dot{\lambda}_m = -\frac{\partial H}{\partial \bar{m}} = -u \frac{\bar{T}_{\max} \lambda_v}{\bar{m}^2}$ is always ≤ 0 . This implies that S_t is negative, and the throttling factor that minimizes the Hamiltonian H is $u^* = 1$.

Finally, the Euler-Lagrange equations, which govern the state and costate dynamics, can be written as:

$$\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\boldsymbol{\lambda}} \end{bmatrix} = \begin{bmatrix} \frac{\partial H}{\partial \mathbf{x}} \\ -\frac{\partial H}{\partial \boldsymbol{\lambda}} \end{bmatrix} = \begin{bmatrix} \bar{\mathbf{v}} \\ -\frac{\bar{\mu}}{\bar{r}^3} \bar{\mathbf{r}} - \frac{\bar{T}_{\max}}{\bar{m}} \boldsymbol{\lambda}_v \\ -\frac{\bar{T}_{\max}}{\bar{I}_{sp} \bar{g}_0} \\ \left(\frac{2k_1(\bar{\rho} - \bar{\rho}_0)}{[k_2 + (\bar{\rho} - \bar{\rho}_0)^2]^2} \right) \bar{\mathbf{r}} - \frac{3\bar{\mu}}{\bar{r}^5} (\bar{\mathbf{r}} \cdot \boldsymbol{\lambda}_v) \bar{\mathbf{r}} + \frac{\bar{\mu}}{\bar{r}^3} \boldsymbol{\lambda}_v \\ -\boldsymbol{\lambda}_r \\ -\frac{\bar{T}_{\max} \lambda_v}{\bar{m}^2} \end{bmatrix}. \quad (53)$$

Since the final time is not fixed, to fully define the Two Point Boundary Value Problem (TP-BVP), it is necessary to impose the transversality condition. In this case, it assumes a particularly simple form since the target state is fixed.

$$H(t_f) = 0 \quad (54)$$

The TPBVP can be solved setting a zero-finding problem that aims to retrieve $\lambda(t_0) = \lambda_0$ and t_f such that the propagated flow $\varphi\left([\bar{\mathbf{x}}_0 \quad \lambda_0]^T, t_0; t_f\right)$ produces $\bar{\mathbf{x}}(t_f)$ and $\lambda(t_f)$ that satisfy the boundary conditions and the transversality condition:

$$\mathbf{F}(\mathbf{y}) = \begin{cases} \bar{\mathbf{r}}(t_f) - \bar{\mathbf{r}}_f = 0 \\ \bar{\mathbf{v}}(t_f) - \bar{\mathbf{v}}_f = 0 \\ \lambda_m(t_f) = 0 \\ H(t_f) = 0 \end{cases} \quad (55)$$

3.4 Optimal solution for $T=3N$

The optimal solution is found with the zero-finding problem described in Algorithm 3. Being the costates quantities with no physical meaning, it was necessary to use random initial guesses. However, knowing that $\dot{\lambda}_m < 0$ and $\lambda_m(t_f) = 0$, it is at least evident that $\lambda_{m,0} \in [0, +250]$. Regarding the final time t_f , the initial guess should be near 20π because the transfer should take more or less 10 orbital periods, so a random value in the interval $[19\pi, 21\pi]$ is chosen.

As the solver options, for the very first iteration **Levenberg-Marquardt** algorithm is used. This algorithm smoothly combines the steepest descent (more stable when far from the solution) and the Gauss-Newton method (faster when near the solution) through a damping parameter, so it is more suitable when random numbers are used. At the end of this iteration, the algorithm is reverted to the default one, which works better when the initial guess is good enough.

Algorithm 3 Zero-finding problem

Require: Initial guess $(\lambda_0^{(0)}, t_f^{(0)})$, N_{max} and $\%_{pert}$

Ensure: Optimal λ_0, t_f

```

1: Initialize  $\mathbf{F}_{opt} = 1, iter = 0$ 
2: while  $flag \neq 1$  and  $iter < N_{max}$  do
3:   if  $iter = 0$  then
4:     Guess  $\lambda_0^{(0)}$  and  $t_f^{(0)}$ 
5:     Solve  $\mathbf{F}(\mathbf{y}) = 0$ 
6:     if  $\|\mathbf{F}\| < \|\mathbf{F}_{opt}\|$  then
7:        $\mathbf{F}_{opt} = \mathbf{F}$ 
8:        $\mathbf{y}_{opt} = \mathbf{y}$ 
9:     end if
10:  else
11:    Perturb previous solution through  $\%_{pert}$ 
12:    Solve  $\mathbf{F}(\mathbf{y}) = 0$ 
13:    if  $\|\mathbf{F}\| < \|\mathbf{F}_{opt}\|$  then
14:       $\mathbf{F}_{opt} = \mathbf{F}$ 
15:       $\mathbf{y}_{opt} = \mathbf{y}$ 
16:    end if
17:  end if
18:   $iter = iter + 1$ 
19: end while
```

Setting options to $\text{FunctionTolerance}=10^{-8}$ and $\text{StepTolerance}=10^{-10}$ were necessary: the first to obtain a precise solution of 4 digits both for position and velocity errors, the second because the function provided to the solver is almost "flat" in that region. In fact, during some iterations, the algorithm could stop because of small step sizes. Eventually $\text{MaxFunctionEvaluations}=10^{-4}$ and $\text{MaxIterations}=5 \cdot 10^3$ are necessary to prevent the solver from being interrupted many times. These stringent tolerances require an accurate propagation as well, so Equation 53 system is integrated using `ode113` with $\text{RelTol}=\text{AbsTol}=3 \cdot 10^{-14}$.

After an intermediate solution is found, it could cause the solver to remain stuck at the local minimum. So, it is necessary to perturb this solution to find a new guess. This is done through the factor $\%_{\text{pert}}$, which allows the selection of new values between random numbers in the neighborhood of the previous solution. The amplitude of this neighborhood is controlled by this percentage of perturbation, which is set to 10%.

The final strategy combines this algorithm with a trial-and-error approach because even after the maximum number of iterations was reached, not always the optimal solution was found. So, it is necessary to restart, provide the last best solution, impose tighter tolerances, and refine by hand the factor $\%_{\text{pert}}$. This procedure was repeated multiple times, continuously reducing the percentage of perturbation, until a solution with an acceptable residual was obtained. This procedure enabled achieving errors much smaller than the specified tolerance, as illustrated in Table 14.

Error	Value	Units
$\ \mathbf{r}(t_f) - \mathbf{r}_f\ $	$2.9024 \cdot 10^{-8}$	km
$\ \mathbf{v}(t_f) - \mathbf{v}_f\ $	$2.9777 \cdot 10^{-8}$	m/s

Table 14: Final state error with respect to target position and velocity ($T_{\text{max}} = 3.000$ N).

Finally, the optimal solution is found (Table 15) and the optimal transfer is propagated as shown in Figure 12 where the gradual change in the orbit inclination is evident. The orbit raising is better shown in the polar plot in Figure 13 where the orbital radius of the transfer is represented on a logarithmic scale, as the initial and final altitudes are very close. The orbit raising is faster at the beginning, confirming that the solution tries to minimize the objective function of debris spatial density which has a monotonic decreasing behavior in the interval $h \in [h_i, h_f] = [700, 1100]$ (Figure 11).

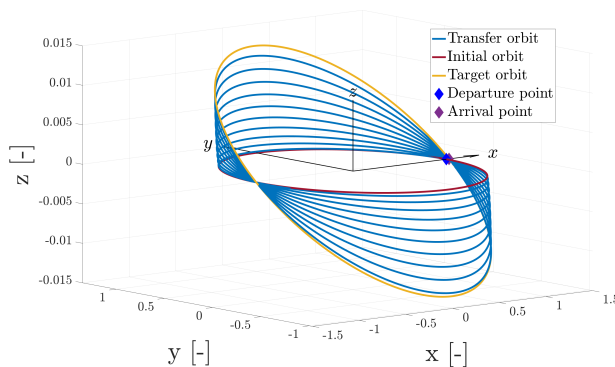


Figure 12: Orbit raising 3D in the Earth-centered equatorial J2000 reference frame

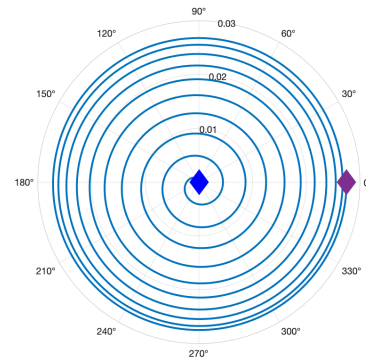
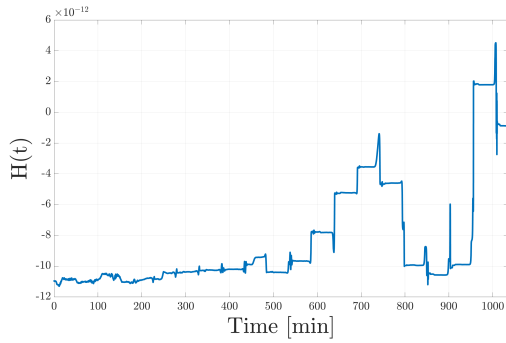
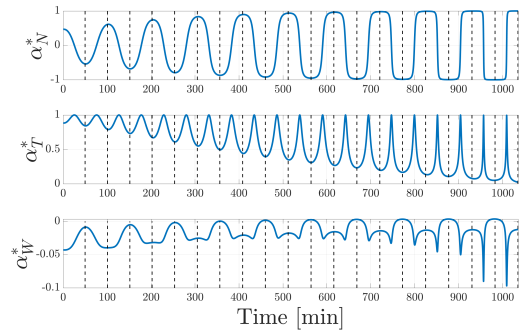


Figure 13: Orbit raising polar plot in logarithmic scale

t_f [mins]		m_f [kg]				
1035.1975		993.9100				
λ_{0,r_x}	λ_{0,r_y}	λ_{0,r_z}	λ_{0,v_x}	λ_{0,v_y}	λ_{0,v_z}	$\lambda_{0,m}$
-214.9802	-10.3659	+0.8856	-10.3929	-214.6095	-112.9434	+2.5964

Table 15: Optimal orbit raising transfer solution ($T_{\max} = 3.000$ N).

Furthermore, to check the solution, the time history of the Hamiltonian function is presented in Figure 14. It can be noted that, except for the numerical errors, it remains near zero. This is a consequence of $H(t) = \text{const}$ for problems in which the objective function and the state dynamics (Equation 48) are not time dependent. So, given the transversality condition in Equation 54, $H(t) = 0$.

**Figure 14:** Hamiltonian time evolution**Figure 15:** Primer vector time evolution

Now, it is possible to retrieve also the evolution in time of the primer vector $\hat{\alpha}^*(t)$ in inertial frame. To show its components in the orthogonal NTW frame, first it is necessary to define these versors in the J2000 inertial frame, for each time step. These versors are the columns of the orthogonal rotation matrix $\mathbf{R}(t)$ that maps a vector from NTW frame to inertial one.

$$\hat{\mathbf{T}}(t) = \frac{\bar{\mathbf{v}}(t)}{\bar{v}(t)} \quad (56)$$

$$\hat{\mathbf{R}}(t) = \frac{\bar{\mathbf{r}}(t)}{\bar{r}(t)} \quad (57)$$

$$\hat{\mathbf{N}}(t) = \hat{\mathbf{R}}(t) \times \hat{\mathbf{T}}(t) \quad (58)$$

$$\hat{\mathbf{W}}(t) = \hat{\mathbf{N}}(t) \times \hat{\mathbf{T}}(t) \quad (59)$$

$$\hat{\mathbf{a}}_{NTW}^*(t) = \mathbf{R}^T(t) \hat{\mathbf{a}}^*(t) \quad (60)$$

The required Primer Vector can now be computed for each time step, and its components are shown in Figure 15. The vertical bars indicate the time instants when the spacecraft crosses the equatorial plane, corresponding to $z = 0$. From the graph of $\hat{\alpha}_N^*$, it can be seen that the maxima and minima coincide with the crossings of the nodal line. This result is logical because the departure and arrival orbits share the same right ascension of the ascending node (RAAN), and the only way to change the orbital plane without altering the RAAN is to perform the plane change maneuver at the nodes. Furthermore, $\hat{\alpha}_T^*$ and $\hat{\alpha}_W^*$ are consistently positive and negative, respectively. This indicates that the in-plane components of $\hat{\alpha}^*$ are always directed towards the velocity and the positive radial direction, which is consistent with an orbit-raising maneuver.

3.5 Optimal solution for lower thrust level

To solve Equation 55 for the thrust level of $T_{max} = 2.860N$ while ensuring fast convergence of the algorithm, numerical continuation was employed. The results for λ_0 and t_f obtained for $T_{max} = 3N$ were used as the initial guess to solve Equation 55. This process was repeated, gradually decreasing the thrust level in 10 intermediate values, until the solution for $T_{max} = 2.860N$ was found. At each step, the solution obtained in the previous step served as the initial guess for the next step. The **OptimalityTolerance** and **FunctionTolerance** were set to 10^{-10} and 10^{-8} , respectively. These stringent values were chosen to limit the propagation of numerical errors during the continuation process. As shown in Table 16, the errors are of the same order of magnitude as those obtained for the previous solution, reported in Table 14.

Error	Value	Units
$\ \mathbf{r}(t_f) - \mathbf{r}_f\ $	$3.1975 \cdot 10^{-7657}$	km
$\ \mathbf{v}(t_f) - \mathbf{v}_f\ $	$3.1947 \cdot 10^{-7}$	m/s

Table 16: Final state error with respect to target position and velocity ($T_{max} = 2.860$ N).

The results obtained are reported in Table 17. It can be noted that the time of flight for $T_{max} = 2.860N$ is approximately $5min$ shorter than that obtained for $T_{max} = 3N$. Consequently, the consumed mass is also smaller.

t_f [mins]	m_f [kg]					
1030.9762	994.2178					
λ_{0,r_x}	λ_{0,r_y}	λ_{0,r_z}	λ_{0,v_x}	λ_{0,v_y}	λ_{0,v_z}	$\lambda_{0,m}$
-593.0491	-11.5902	+2.2970	-11.9291	-592.6852	-920.1372	+17.3771

Table 17: Optimal orbit raising transfer solution ($T_{max} = 2.860$ N).

Now, Figure 16 shows that the transfer is more homogeneous than the previous case in Figure 13. Additionally, the propagation of the numerical error is more evident when Figure 14 and Figure 17 are compared. In fact, there is an ≈ 2 order of magnitudes difference. In any case, the value of the Hamiltonian function remains again very close to zero.

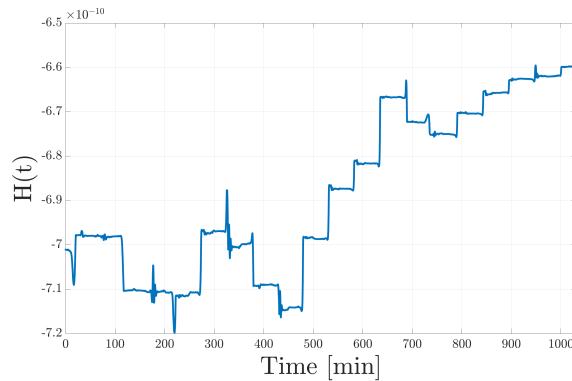
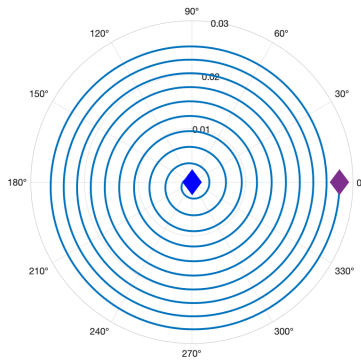


Figure 16: Orbit raising polar plot for lower thrust **Figure 17:** Hamiltonian time evolution for lower thrust

Finally, Figure 18 confirms the more regular behavior of the transfer in terms of components of the Primer Vector $\hat{\alpha}^*$. However, it can still be observed that $\hat{\alpha}_T^*$ and $\hat{\alpha}_N^*$ have slightly higher magnitudes during the initial phase of the transfer. This behavior is consistent with the shape of the debris density distribution shown in Figure 11. Furthermore, the plane change is performed

primarily around the nodes, and the signs of the in-plane components of $\hat{\alpha}^*$ are consistent with an orbit-raising maneuver.

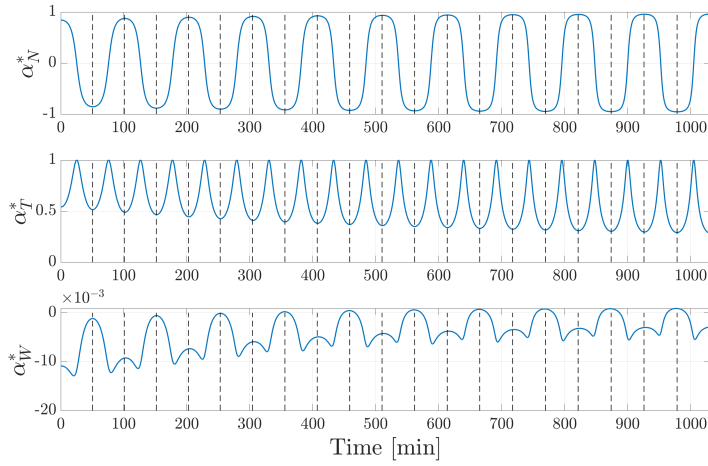


Figure 18: Primer vector time evolution for lower thrust

References

- [1] Francesco Topputo. On optimal two-impulse earth–moon transfers in a four-body model. *Celestial Mechanics and Dynamical Astronomy*, 117:279–313, 2013.

A Lagrangian Points

The Triangular points L_4 and L_5 can also be computed by hand, solving the intersection of the two unitary circles centered in the Earth and Moon. Knowing that the equilibrium points lie on the plane of the two bodies, the L_4 and L_5 coordinates are retrieved as follows:

$$L_4 = \left[1 - \mu - \frac{1}{2}; \sqrt{1 - \left(\frac{1}{2}\right)^2}; 0 \right] \quad L_5 = \left[1 - \mu - \frac{1}{2}; -\sqrt{1 - \left(\frac{1}{2}\right)^2}; 0 \right] \quad (61)$$

With regard to the Collinear points, their coordinates can be found as the intersection of $\frac{\partial \Omega}{\partial x}(x, y)$ with the x -axis through the function **fzero**.

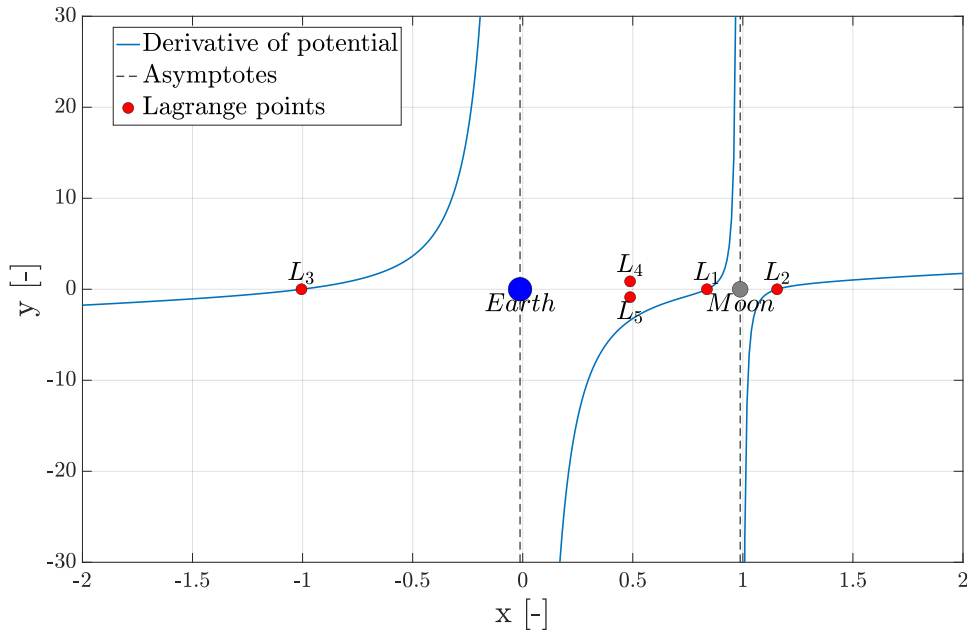


Figure 19: Lagrangian points in Earth–Moon adimensional rotating frame

Figure 19 highlights that the function $\frac{\partial \Omega}{\partial x}(x, y)$ presents three regions divided by the two asymptotes $x = -\mu$ and $x = 1 - \mu$, which correspond to the positions of the two attractors (Earth and Moon). Thus, the function **fzero** has to be used three times to search the solutions in these three regions.

B CRTBP dynamics

Equations of motion of a spacecraft in the Circular Restricted Three-Body Problem (CRTBP) in the form of $\dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x})$:

$$\begin{cases} \dot{x} = v_x \\ \dot{y} = v_y \\ \dot{z} = v_z \\ v_x = 2v_y + \frac{\partial \Omega}{\partial x}(x, y, z) = 2v_y + x - \frac{(1-\mu)(x+\mu)}{r_1^3} - \frac{\mu(x+\mu-1)}{r_2^3} \\ v_y = -2v_x + \frac{\partial \Omega}{\partial y}(x, y, z) = -2v_x - \frac{y(1-\mu)}{r_1^3} - \frac{\mu y}{r_2^3} \\ v_z = \frac{\partial \Omega}{\partial z}(x, y, z) = -\frac{z(1-\mu)}{r_1^3} - \frac{\mu z}{r_2^3} \end{cases} \quad (62)$$

C PBRFBP dynamics

Definition of the Planar Bicircular Restricted Four-Body Problem potential Ω_4 starting from the CRTBP potential Ω_3 :

$$\Omega_4(x, y) = \Omega_3(x, y) + \frac{m_s}{r_3(t)} - \frac{m_s}{\rho^2}(x \cos(\omega_s t) + y \sin(\omega_s t)) \quad (63)$$

where

$$r_3(t) = \sqrt{(x - \rho \cos(\omega_s t))^2 + (y - \rho \sin(\omega_s t))^2} \quad (64)$$

The resulting equation of motions in the form of $\dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x})$:

$$\begin{cases} \dot{x} = v_x \\ \dot{y} = v_y \\ v_x = 2v_y + \frac{\partial \Omega_4}{\partial x}(x, y, z) \\ v_y = -2v_x + \frac{\partial \Omega_4}{\partial y}(x, y, z) \end{cases} \quad (65)$$

where

$$\begin{aligned} \frac{\partial \Omega_4}{\partial x} &= x + \frac{(2\mu + 2x)(\mu - 1)}{2r_1^3} - \frac{\mu(\mu + x - 1)}{r_2^3} - \frac{m_s(x - \rho \cos(\omega_s t))}{r_3^3} - \frac{m_s \cos(\omega_s t)}{\rho^2} \\ \frac{\partial \Omega_4}{\partial y} &= y + \frac{y(\mu - 1)}{r_1^3} - \frac{\mu y}{r_2^3} - \frac{m_s(y - \rho \sin(\omega_s t))}{r_3^3} - \frac{m_s \sin(\omega_s t)}{\rho^2} \end{aligned} \quad (66)$$

D NLP formulation

Non-Linear Programming problem is stated as:

$$\min_{\mathbf{y}} f_{obj}(\mathbf{y}) \quad \text{such that:} \quad \begin{cases} \mathbf{g}(\mathbf{y}) \leq 0 \\ \mathbf{h}(\mathbf{y}) = 0 \end{cases} \quad (67)$$

Inequality and equality constraint vectors $\mathbf{g}(\mathbf{y})$ and $\mathbf{h}(\mathbf{y})$ can be divided into linear constraints $\mathbf{b}$, nonlinear constraints $\mathbf{c}$, and lower/upper bounds to allow the correct use of the Matlab function `fmincon`.

$$\min_{\mathbf{y}} f_{obj}(\mathbf{y}) \quad \text{such that:} \quad \begin{cases} A_{eq}\mathbf{y} = \mathbf{b}_{eq} \\ A_{ineq}\mathbf{y} \leq \mathbf{b}_{ineq} \\ \mathbf{c}_{eq}(\mathbf{y}) = 0 \\ \mathbf{c}_{ineq}(\mathbf{y}) \leq 0 \\ \mathbf{y}^{LB} \leq \mathbf{y} \leq \mathbf{y}^{UB} \end{cases} \quad (68)$$

E Jacobian matrix of PBRFBP

The jacobian $A(t)$ of the RHS of the PBRFBP is here reported.

$$A(t) = \frac{\partial \mathbf{f}}{\partial \mathbf{x}} = \begin{bmatrix} \mathbf{0}_{2 \times 2} & \mathbf{I}_{2 \times 2} \\ \begin{bmatrix} \frac{\partial^2 \Omega}{\partial x^2} & \frac{\partial^2 \Omega}{\partial x \partial y} \\ \frac{\partial^2 \Omega}{\partial y \partial x} & \frac{\partial^2 \Omega}{\partial y^2} \end{bmatrix} & \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} \end{bmatrix} \quad (69)$$

where

$$\frac{\partial^2 \Omega}{\partial x^2} = 1 - \frac{(1-\mu)}{r_1^3} + \frac{3(1-\mu)(x+\mu)^2}{r_1^5} - \frac{\mu}{r_2^3} + \frac{3\mu(x+\mu-1)^2}{r_2^5} - \frac{m_s}{r_3^3} + \frac{3m_s(x-\rho \cos(\omega_s t))^2}{r_3^5} \quad (70)$$

$$\frac{\partial^2 \Omega}{\partial y^2} = 1 - \frac{(1-\mu)}{r_1^3} + \frac{3(1-\mu)y^2}{r_1^5} - \frac{\mu}{r_2^3} + \frac{3\mu y^2}{r_2^5} - \frac{m_s}{r_3^3} + \frac{3m_s(y-\rho \sin(\omega_s t))^2}{r_3^5} \quad (71)$$

$$\frac{\partial^2 \Omega}{\partial x \partial y} = \frac{\partial^2 \Omega}{\partial y \partial x} = \frac{3(1-\mu)(x+\mu)y}{r_1^5} - \frac{3\mu(x+\mu-1)y}{r_2^5} + \frac{3m_s(x-\rho \cos(\omega_s t))(y-\rho \sin(\omega_s t))}{r_3^5} \quad (72)$$

F Intermediate multiple shooting solution

With the very same options adopted in the simple shooting (constraint tolerance of 10^{-10}), the solver is able to find just an intermediate solution, as the maximum default number of iterations of $4 \cdot 10^2$ is reached. This justifies the fact that, to reach the solution, a large number of iterations and function evaluations are needed. For the sake of completeness, here in Figure 20 the intermediate solution is reported.

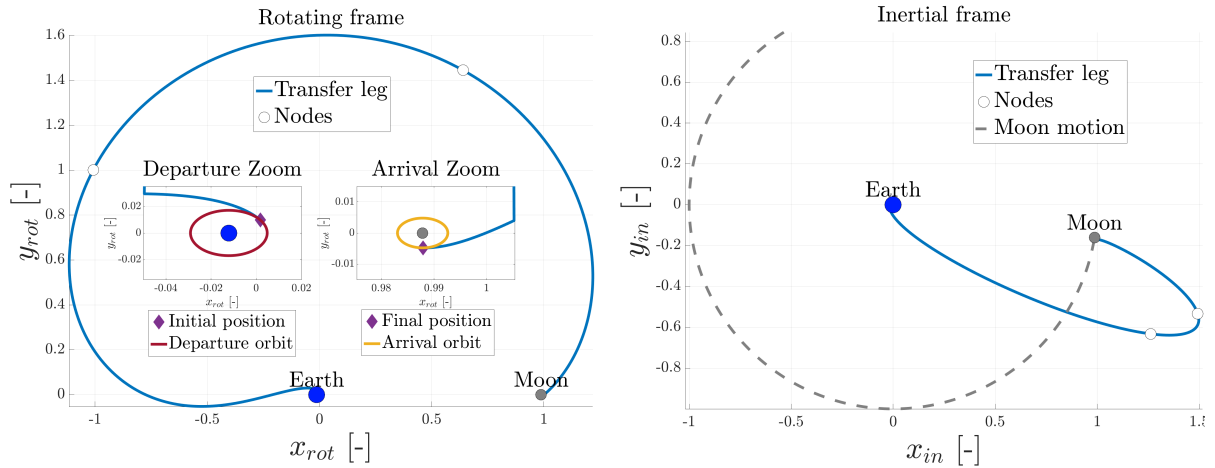


Figure 20: Intermediate multiple shooting solution in both adimensional frames

The solution found after **400** iterations and **1285** function evaluations, with a maximum constraint error of $\mathbf{1.006536 \cdot 10^{-5}}$, results in $\Delta v = \mathbf{4.006351[-]} = \mathbf{4.104667 km/s}$ and $\Delta t \approx \mathbf{17.306 days}$. From Figure 20 it can be noted that the shape of trajectory is similar to the one obtained with the simple shooting, along with the results in terms of Δv and Δt . The obtained results in terms of initial condition, t_i and t_f are presented in Table 18.

$r_{x,0}$ [DU]	$r_{y,0}$ [DU]	$v_{x,0}$ [VU]	$v_{y,0}$ [VU]	t_i [TU]	t_f [TU]
+0.001816	+0.009739	-6.118259	+8.774328	2.136	6.121

Table 18: Intermediate multiple shooting solution in the Earth-Moon rotating frame.